

Amministrazione di Sistema

Virtualizzazione

Virtualizzazione

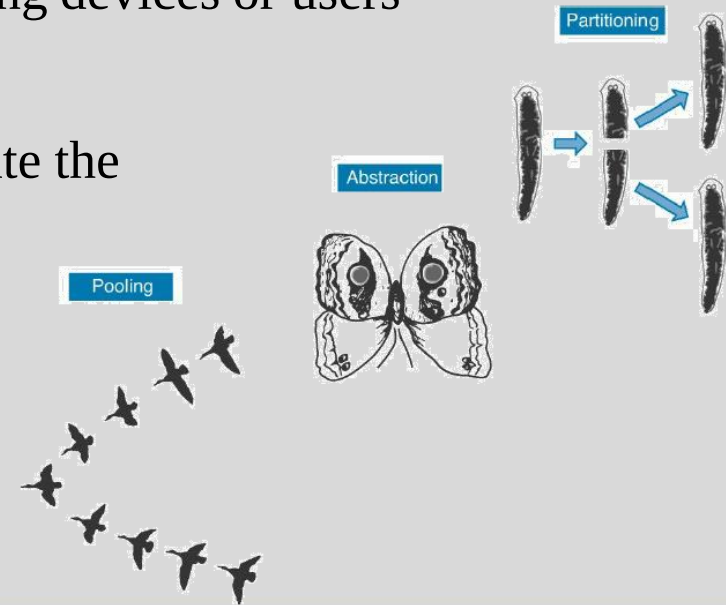
- Definizione (fonte: Wikipedia 2024)
 - possibilità di astrarre le componenti hardware, cioè fisiche, degli elaboratori al fine di renderle disponibili al software in forma di risorsa virtuale
 - definizione di virtuale
 - “oggetto non realmente esistente, non avente una sua fisicità”
- Definizione (fonte: Oxford English Dictionary)
 - in the context of computing, virtual means “not physically existing as such but made by software to appear to do so”
 - in the same context, it can be said that a virtual element is a particular abstraction of such an element
- Since the early days of the Internet boom in the 1990s, it has become a complete cliché (or simply lazy writing) to name any web-related activity as “virtual”

Virtualizzazione

- Virtualizzazione
 - livello di astrazione per creare contesti multipli, partizionando le risorse
 - abstraction is the process of considering something independently of its associations or attributes
- Qualunque risorsa hardware o software può essere virtualizzata
 - sistemi operativi, server, processori, memoria, spazio disco, applicazioni, fax, telefoni, ...
 - virtualization is not restricted to a single technology area in the data center

Tipi di virtualizzazione

- Pooling
 - several physical elements work simultaneously to form a single logical entity that shares characteristics with the original entities
- Abstraction
 - One physical resource creates a single virtual element to offer different characteristics to interacting devices or users
- Partitioning
 - independent logical partitions that emulate the characteristics of the physical resource



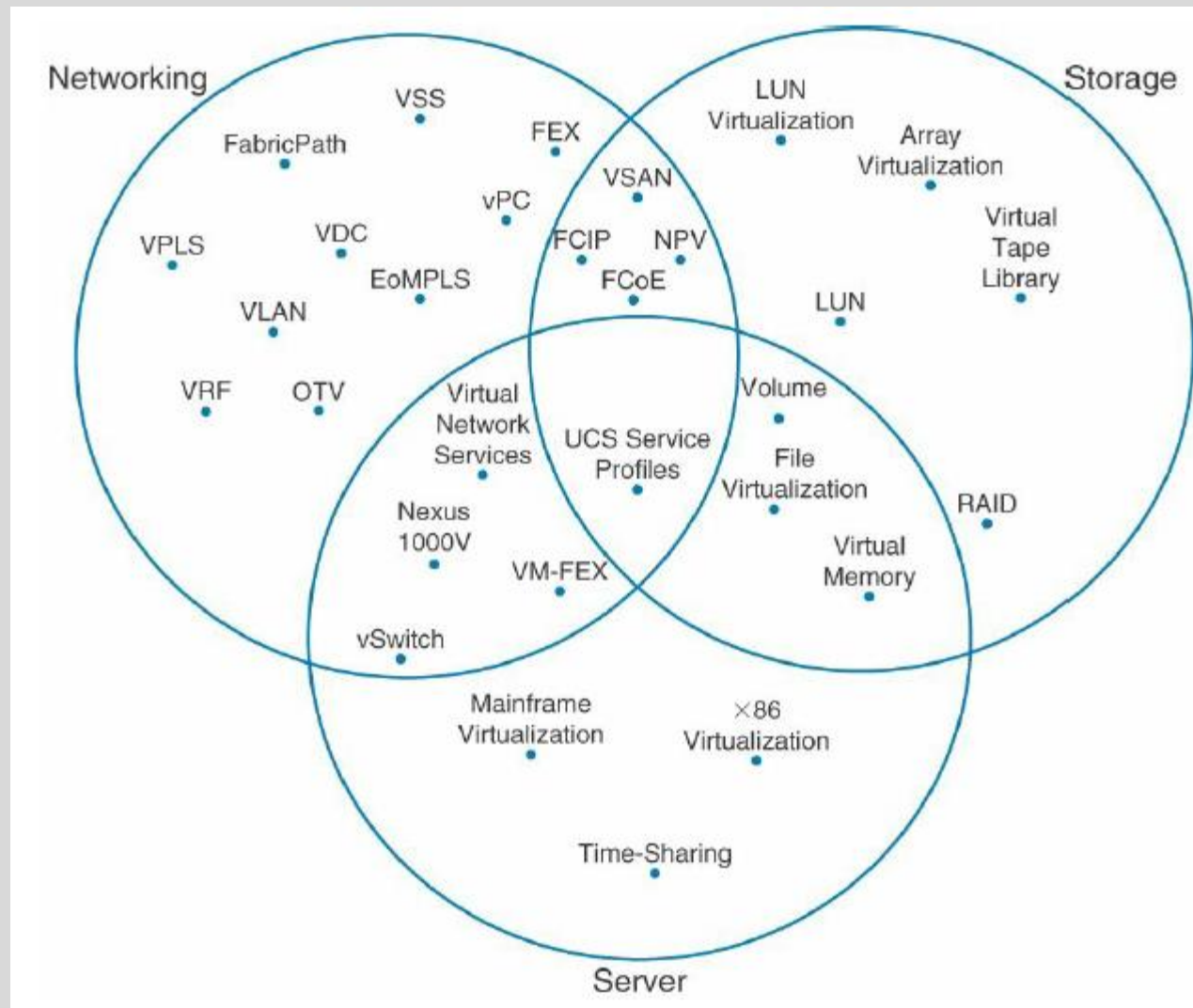
Categorie di virtualizzazione (1)

- System Virtualization
 - Server Virtualization
 - Desktop Virtualization
 - Virtualizzazione lato client
 - Sicurezza, integrazione, trasparenza?
 - Virtualizzazione lato server
 - VDI (Virtual Desktop Infrastructure)
- Application Virtualization
 - “Streaming” di applicazioni
 - Microsoft Application Virtualization App-V (SoftGrid)
 - Applicazioni “portabili”
 - Portable application, “Bolle applicative”, VMware ThinApp
 - Gestione multi-layer

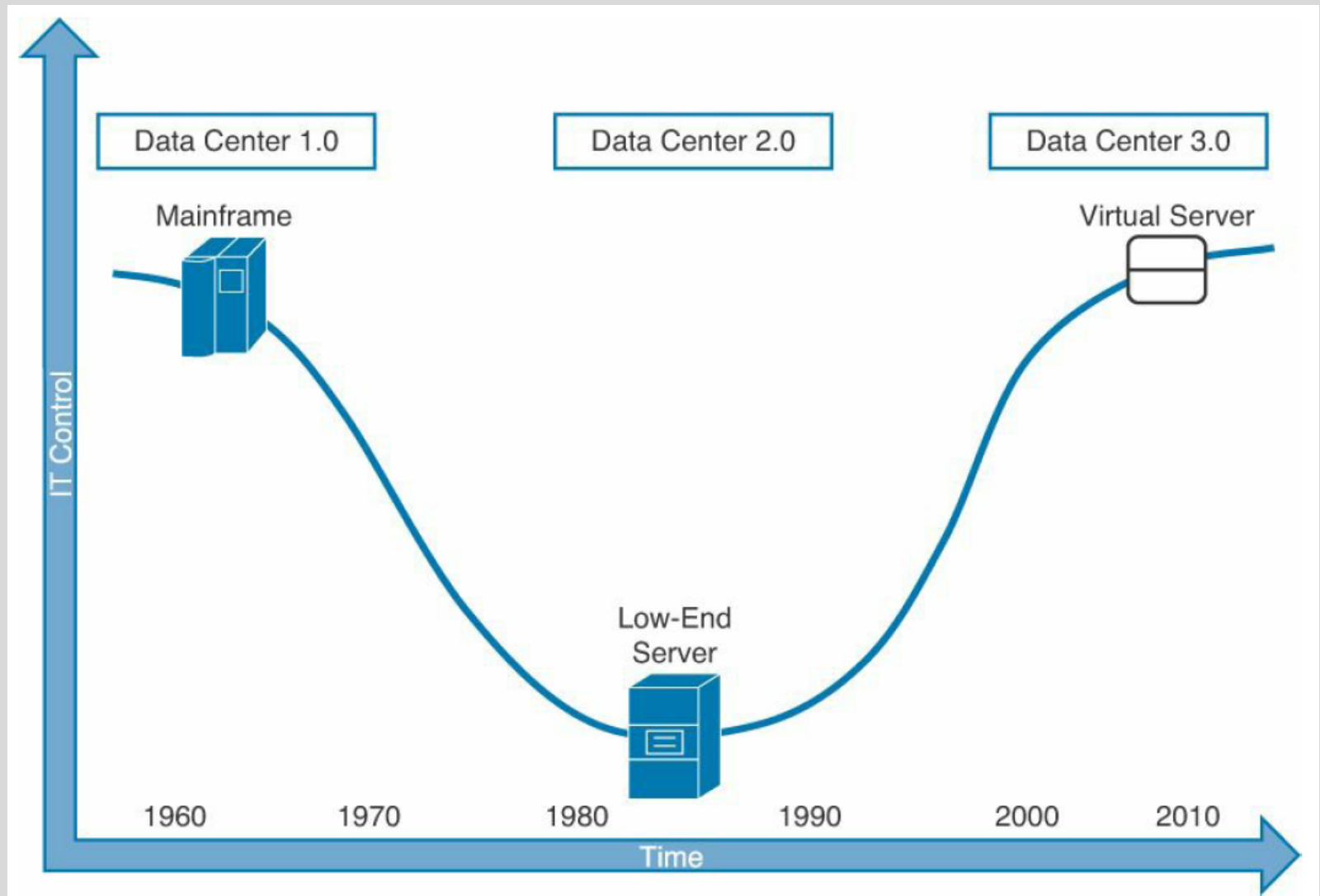
Categorie di virtualizzazione (2)

- Presentation Virtualization
 - Microsoft RDS, Citrix XenApp/XenDesktop
- User State Virtualization
 - Document redirection / Offline file
- Storage Virtualization
 - SAN, SDS
- LAN Virtualization
 - VLAN, VPN
 - Network Virtualization, SD-WAN, SDN, NVF
- Data Virtualization
- Altri esempi di «virtualizzazione»
 - Cloud, Cluster, ...

Esempi di virtualizzazione



Evoluzione del datacenter



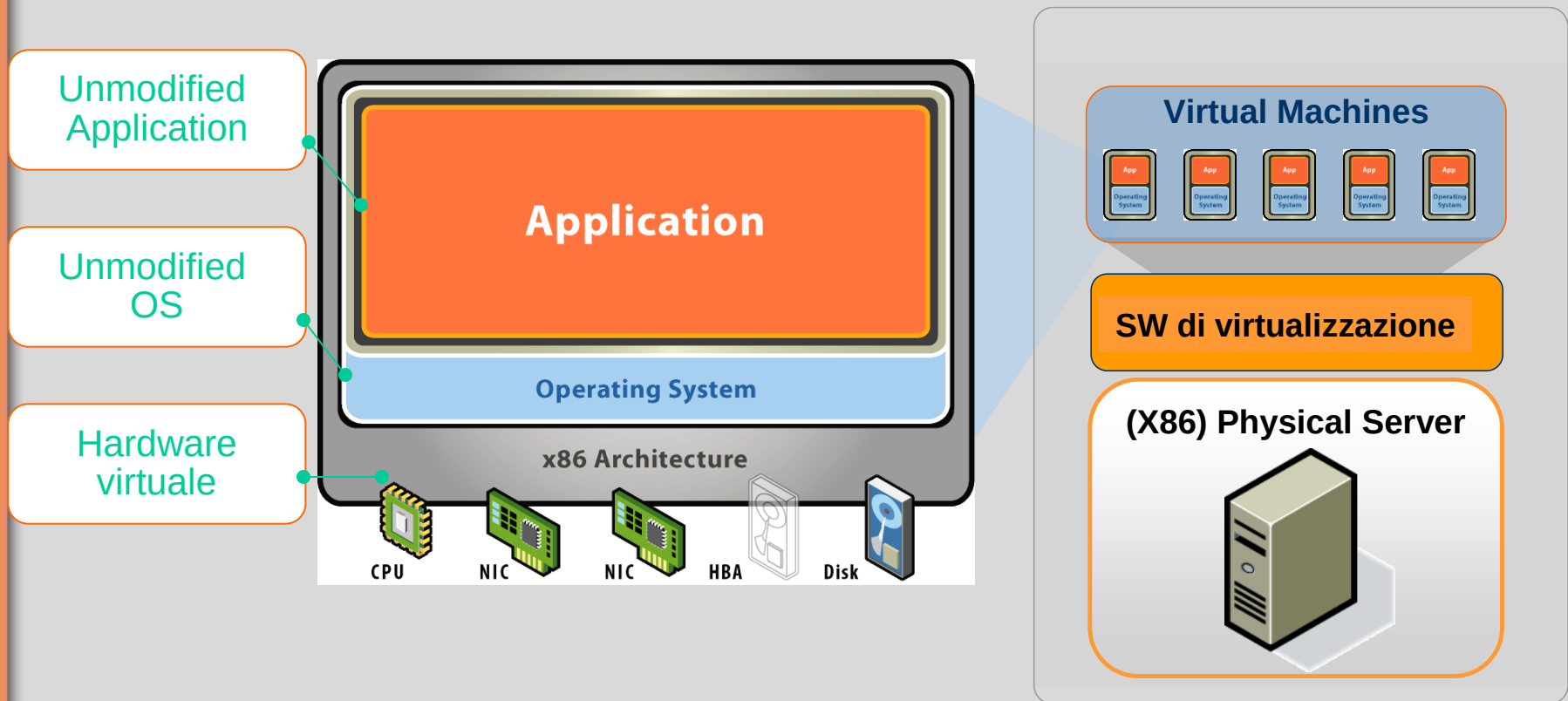
Virtualizzazione

Virtualizzazione Server

Cenni storici

- La preistoria (anni '60)
 - nasce il concetto di macchina virtuale
 - da parte di IBM con il suo prodotto VM/370
 - si parla di SuperVisor
- La storia (anni '90)
 - la ricerca di un meccanismo per rendere un programma portabile su più architetture ha portato nuovo interesse alle tecniche di virtualizzazione
 - macchina virtuale simulata: Java Virtual Machine (JVM)
 - negli anni recenti c'è stato un rinnovato interesse per la virtualizzazione dell'hardware
 - 1999 – primi prodotti di virtualizzazione x86

Sistema virtuale



Terminologia

- Virtualization / Paravirtualization
 - Desktop virtualization
 - Server virtualization
- Host / Parent / Domain0
 - Macchina reale (di solito fisica) sulla quale viene eseguito il software di virtualizzazione
- Guest / VM / DomainUs / istanza logica
 - La macchina virtuale
- Hypervisor
 - VMM (Virtual Machine Monitor)
- Processi di conversione
 - P2V (Physical To Virtual)
 - V2V (Virtual To Virtual)

Key Features of Virtualization

Partitioning



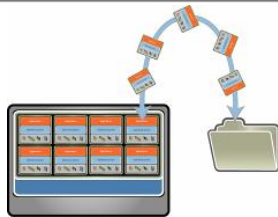
Run multiple virtual machines simultaneously on a single physical server

Isolation



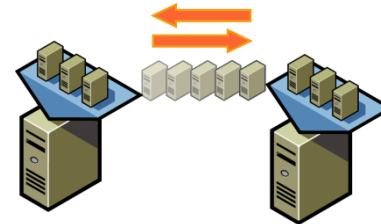
Each virtual machine is isolated from other virtual machines on the same server

Encapsulation



Entire virtual machine is saved in files and can be moved and copied by moving and copying files

Hardware Independence



Run a virtual machine on any server without modification

Consolidation and Containment

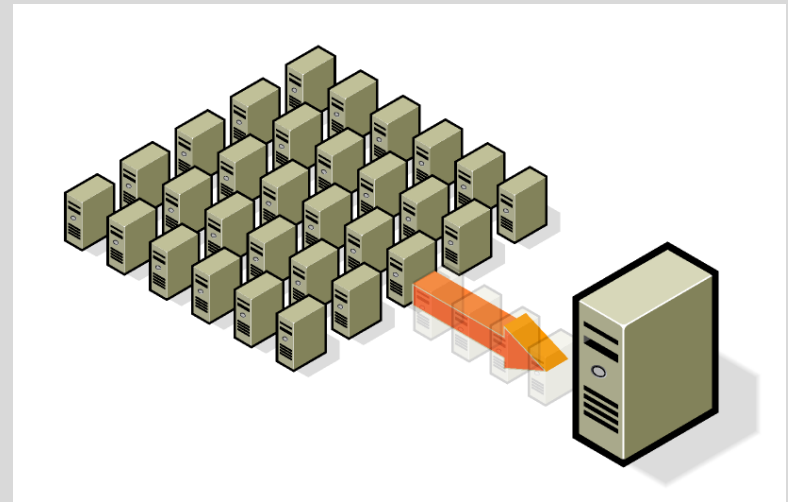
Virtual infrastructure provides a managed approach to regain control over x86-based server sprawl

Server consolidation

- Reduce number of servers in datacenter to reduce costs

Server containment

- Create virtual machines instead of provisioning new hardware
- Reduces future hardware needs

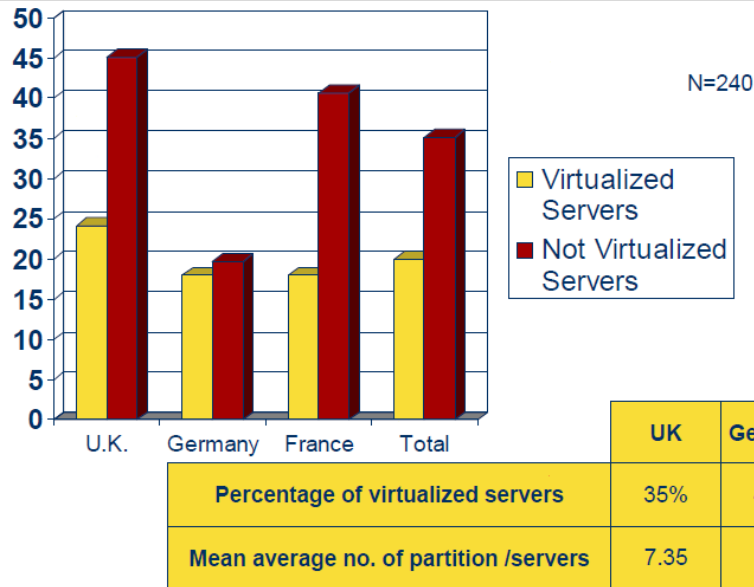


Aspetto ecologico-energetico: risparmio di energia e minor inquinamento

ROI (Return On Investment)

- Le soluzioni della virtualizzazione riducono i costi
 - acquisto dell'hardware
 - allocazione fisica
 - manutenzione
 - amministrazione centralizzata
 - espansione del datacenter e provisioning di nuovi server
 - test
 - aggiornamento dell'hardware
- Stime e analisi indicative
 - ritorno completo sull'investimento in meno di 6 mesi
 - riduzione del TCO del Datacenter fino al 64 %
 - riduzione dei costi hardware del 40%
 - riduzione dei costi di manutenzione e testing fino al 70%
 - riduzione dei tempi di downtime del 25-55 %
 - riduzione dei tempi di failure recovery dell' 80%
 - riduzione dei tempi di sviluppo e test da ore a minuti

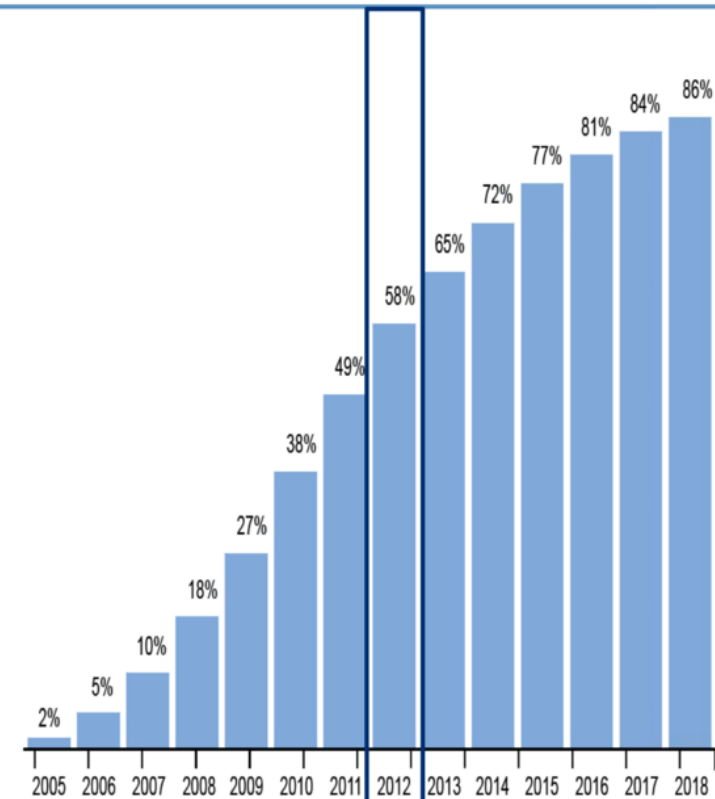
Server fisici vs. Server virtuali



IDC 2009

- Nel 2009 ci doveva essere il sorpasso
- In ogni caso la virtualizzazione server è diventata mainstream

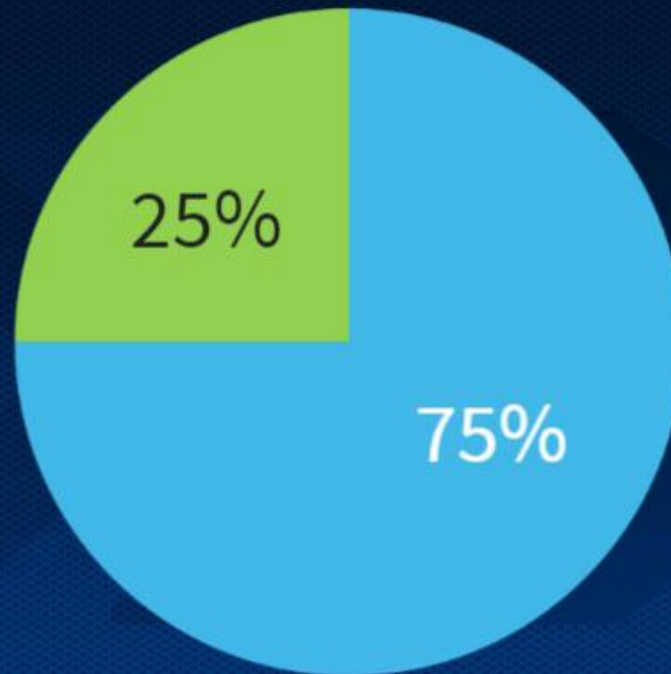
Figure 1. Percentage of x86-Architecture Workloads Running in VMs



Source: Gartner (March 2011)

Server fisici vs. Server virtuali

Non-virtualized



75%

Virtualized

May 2016*

*Gartner Says Worldwide Server Virtualization Market Is Reaching Its Peak <http://www.gartner.com/newsroom/id/3315817>

Virtual vs. Real

- Vantaggi
 - Consolidamento server
 - Ottimizzazione delle risorse
 - processore, disco, memoria, ...
 - Ottimizzazione degli spazi
 - alternativa ai server Blade
- Svantaggi
 - Criticità da gestire?
 - Perdita di efficienza?
- Considerazioni sulle licenze
 - Ogni sistema (host o virtuale) necessita di licenze
 - <http://www.microsoft.com/licensing/highlights/virtualization/faq.mspx>

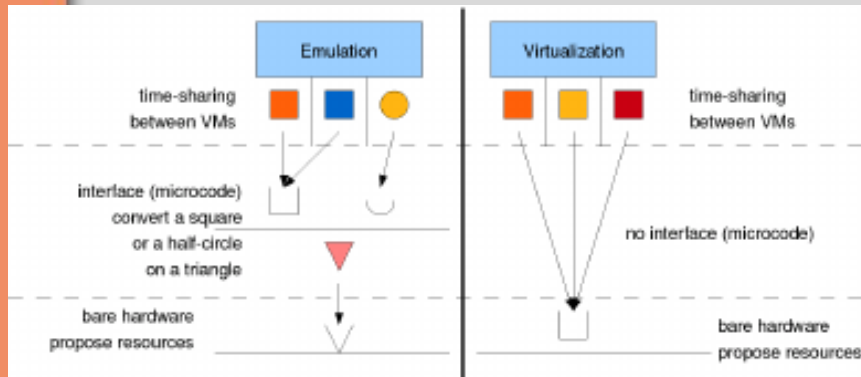
Prestazioni

- Hypervisor Tax
 - Running a hypervisor comes with a “tax” on performance
 - Overhead usually less than 5% on the physical server resources
- Noisy Neighbors
 - VMs can “share” the physical resources of a bare metal server
 - If a neighboring VM runs a resource-heavy application, this can impact the performance of your VM as more of the communal resources are redirected
- I/O blender effect
 - Unhealthy I/O: Small, Fractured, Random
- VM Sprawl

Tipi di virtualizzazione

- Partizionamento hardware
- Virtualizzazione
 - meccanismo per creare contesti multipli su di un singolo processore, partizionando le risorse
 - Virtualizzazione completa (Full)
 - convenienza?
 - Paravirtualizzazione
 - non è richiesta l'emulazione dei driver (in particolare quelli di I/O)
 - l'overhead è molto basso
 - OS level virtualization
- Emulazione
 - eseguire S.O. e applicazioni compilate per una architettura su di un'altra
- Simulazione

Emulazione <> Virtualizzazione



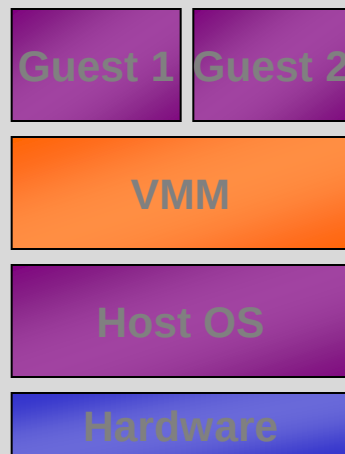
VIRTUALIZATION	EMULATION
In virtualization, hardware can be accessed directly.	In case of Emulation, you'd need a software connector to access hardware.
Virtual machine can run the code directly, which is available in different languages.	Emulator requires an interpreter to translate the source code.
Virtual machines are relatively faster in its operations.	Emulators are relatively slower.
VM solutions are costlier than Emulation.	Emulation is comparatively cheaper.
Virtualization provides better backup solutions.	Emulation falls short of virtualization as far as backup and recovery is considered.

Livelli e virtualizzazione

- Da quale livello viene introdotta la virtualizzazione?
 - Diversi modelli di hypervisor
- Quali livelli vengono virtualizzati
 - Dal livello «virtualizzazione»
 - Nested virtualization
 - Dal livello hardware
 - Virtualizzazione full
 - Emulazione
 - Dal livello del sistema operativo
 - Dal livello applicativo

Modelli di hypervisor

Type-2 VMM

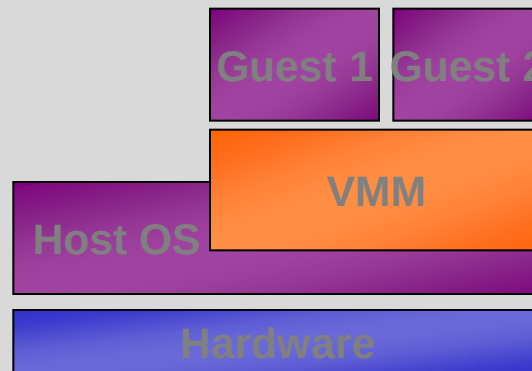


Esempi:

JVM

CLR

**Hybrid VMM
(Type-2 evoluto)**



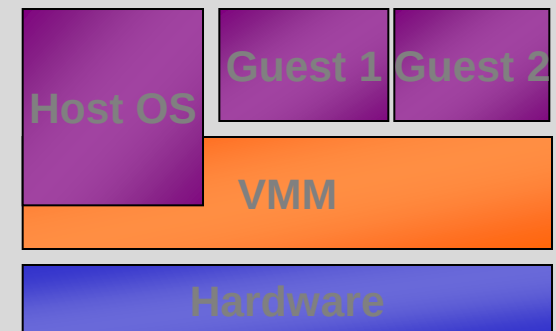
Esempi:

Microsoft Virtual PC

VMware Workstation

VirtualBox

**Type-1 VMM
(Hypervisor)**



Esempi:

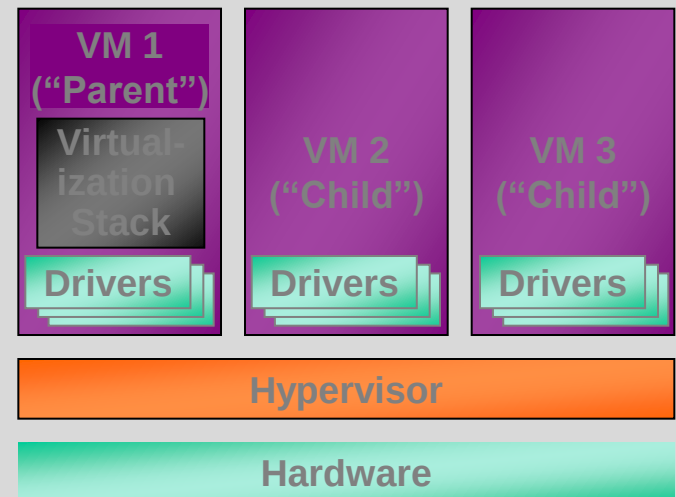
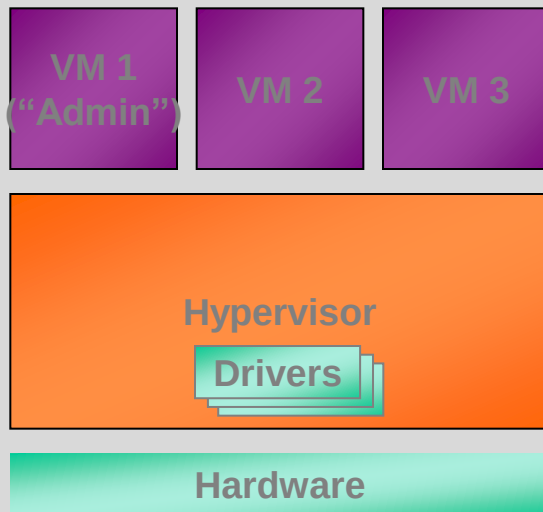
VMware ESX/ESXi

Microsoft Hyper-V

Xen / KVM

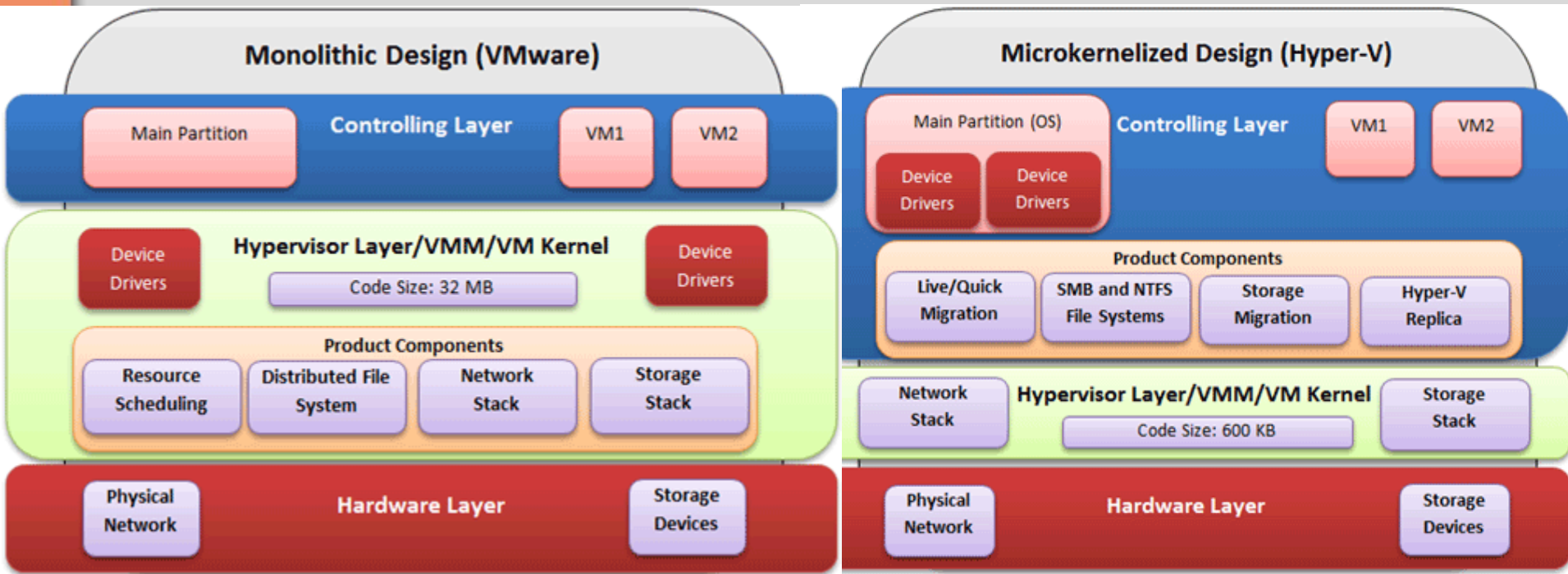
Tipi di hypervisor (type 1)

- Monolithic hypervisor
 - Simpler than a modern kernel, but still complex
 - Contains its own drivers' model
- Microkernelized hypervisor
 - Simple partitioning functionality
 - Increase reliability and minimize TCB
 - No third-party code
 - Drivers run within guests



Microkernelized Hypervisor has an inherently secure architecture with minimal attack surface

Monolítico vs. Microkernel



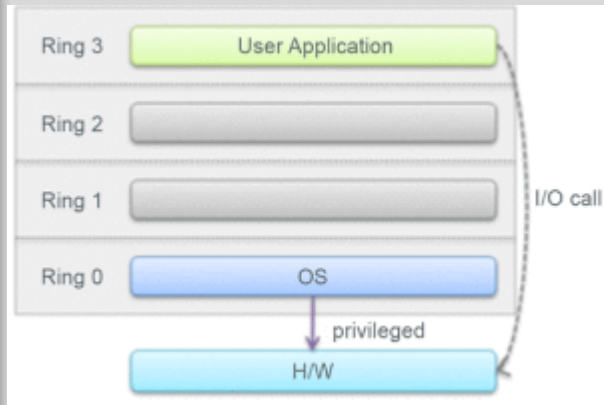
Virtualizability

- Problema delle Sensitive Instructions
 - Goldberg. Architectural Principles for Virtual Computer Systems. Ph.D. thesis, Harvard University, 1972
 - Mode Referencing Instructions
 - Sensitive Register/Memory Access Instructions
 - Storage Protection System Referencing Instructions
 - All I/O Instructions
- Teorema relativo alla strict virtualizability
 - Popek and Goldberg, Formal requirements for virtualizable third generation architectures - SOSPP '73
 - *For any conventional third generation computer, a virtual machine monitor may be constructed if the set of sensitive instructions for that computer is a subset of the set of privileged instructions*

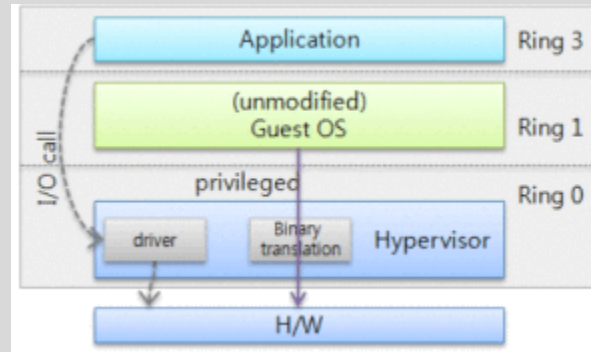
Diversi modelli

- Trap and Emulate
 - Let VM execute most of its instructions directly on the h/w
 - Except for some sensitive instructions that trap into the VMM and are emulated
- Paravirtualizzazione
 - occorre modificare il kernel del sistema guest
- Virtualizzazione Full
 - Assisista in hardware

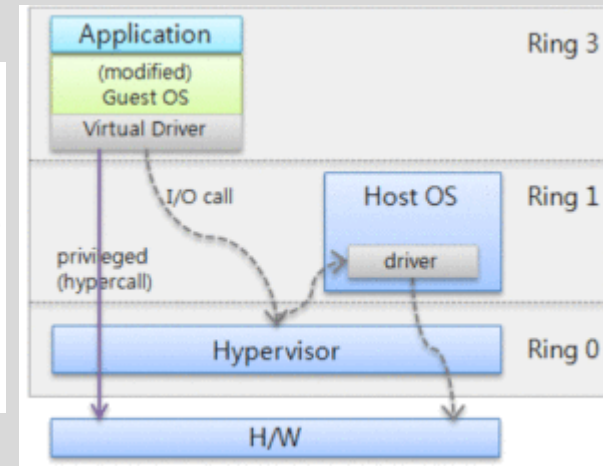
Ring di protezione



Sistema
fisico

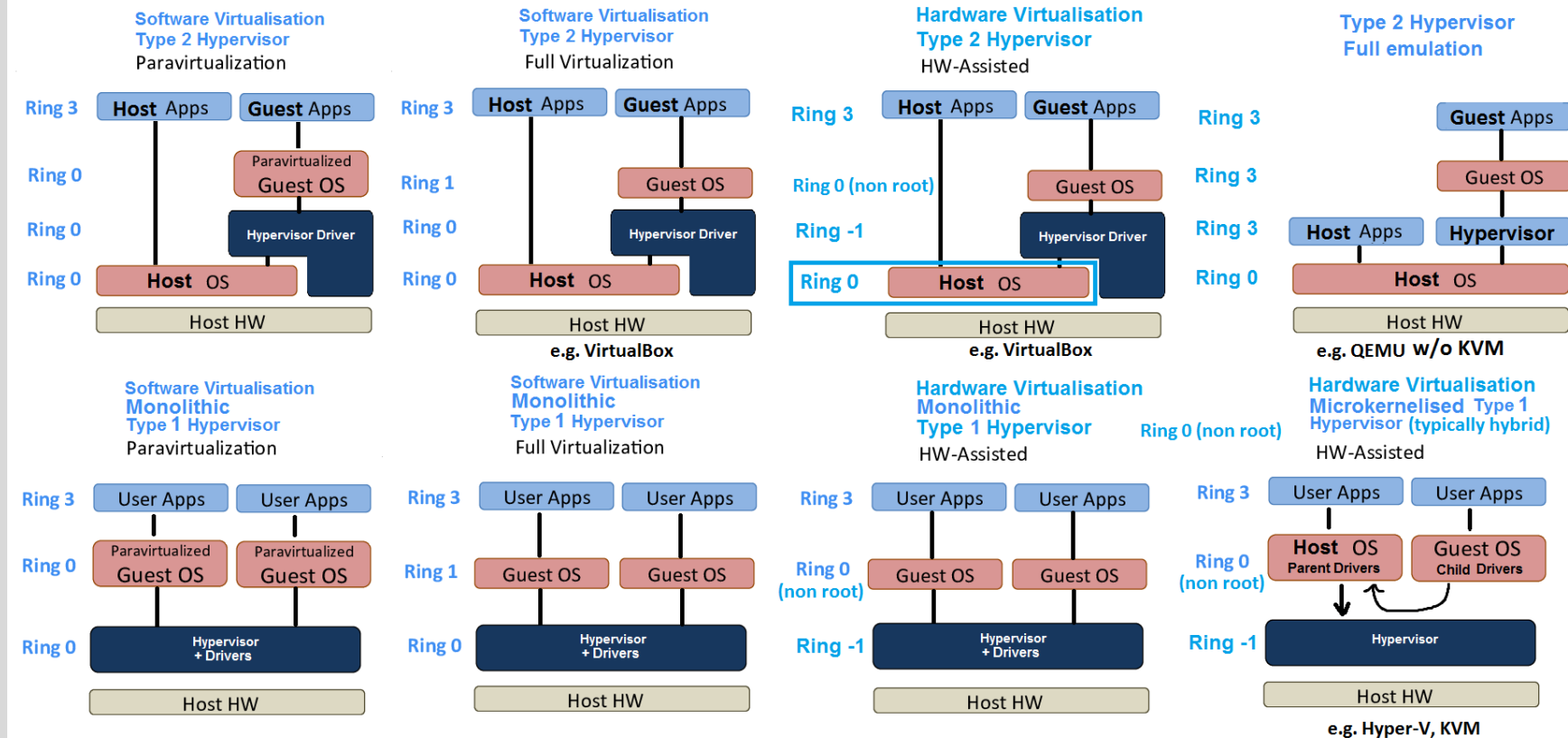


Virtualizzazione
Full



Paravirtualizzazione

Tassonomia completa



Virtualizzazione delle risorse

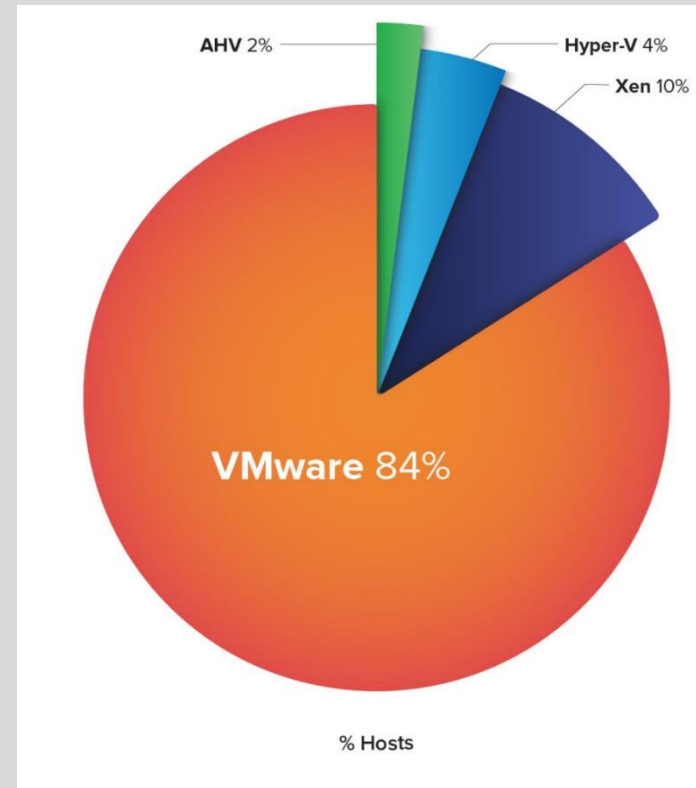
- CPU
 - Virtualizzazione Full
 - 1 vCPU = 1 pCore
 - Partizionamento con possibilità di over-provisioning
 - Condivisione di risorse (rapporto < 4:1)
- RAM/PMEM
 - Virtualizzazione Full
- Disk
 - Emulazione o Paravirtualizzazione
 - 1 disco virtuale = 1 file
- Network
 - Emulazione o Paravirtualizzazione
 - Introduzione dei virtual switch e delle vNIC
- GPU

Prodotti di virtualizzazione (1)

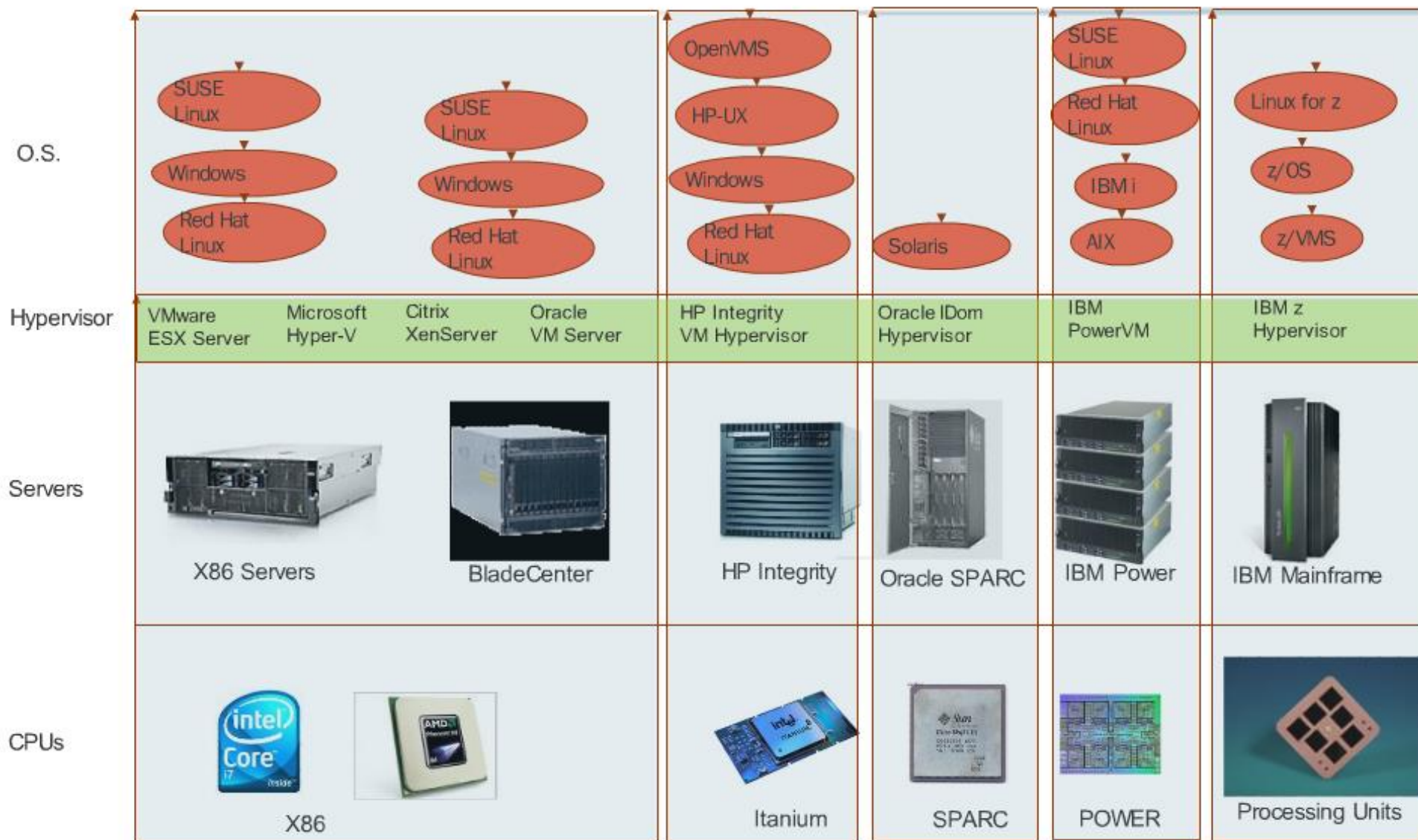
- Uso Desktop lato client
 - Software proprietari gratuiti
 - VMware Workstation Player
 - ~~Microsoft Virtual PC~~ / Microsoft Hyper-V (Win10/11)
 - Software proprietari a pagamento
 - VMware Workstation Pro / Fusion
 - Parallels
 - Software libero
 - Qemu
 - VirtualBox OSE
 - Oracle/Sun VirtualBox
 - multiplatforma
 - Emulatori di altri sistemi

Prodotti di virtualizzazione (2)

- Uso Server
 - Software libero o OpenSource
 - KVM
 - XEN
 - (V-server, UML, OpenVZ)
 - Software proprietari gratuiti
 - ~~Microsoft Virtual Server 2005~~
 - Microsoft Hyper-V “Viridian”
 - ~~VMware Server (ex GSX)~~
 - Software proprietari a pagamento
 - VMware vSphere (ESXi + vCenter)
 - http://www.vmware.com/products/server/esx_features.html
 - Citrix Xen Enterprise
 - Proxmox
 - Virtuozzo Server Virtualization / Parallels
 - Nutanix AHV



Virtualizzazione e architetture



Soluzioni hardware

– IBM

- Logical Partitions (LPAR – AIX 5.1)
- Dynamic Logical Partitions (DLPAR – AIX 5.2)
- Shared Processor Logical Partitions (SPLPAR – AIX 5.3 on Power 5)

– HP

- Hardware partitions (nPar)
- Virtual partitions (vPar)

– Sun/Oracle

- LDOMs
- Dynamic System Domains (DSD)
- Containers (Zones - Solaris 10)

– Intel

- Hyper-threading
- Virtualization Technology (VT)

Intel Virtualization Technology

Intel® VT Roadmap: Overview

Vector 3:
I/O Focus

PCI-SIG

Standards for I/O-device sharing:

- Natively sharable I/O devices
- Endpoint DMA-translation caching

Vector 2:
Platform Focus

VT-d

Infrastructure for I/O-device virtualization:

- DMA protection and remapping
- Interrupt filtering and remapping

Vector 1:
Processor Focus

VT-x

VT-i

Establish foundation for virtualization in the Intel® 64 and Itanium® architectures...

... followed by on going evolution of support:

- Microarchitectural (e.g., lower VM entry/exit costs)
- Architectural (e.g., extended page tables – EPT)

VMM
Software
Evolution

Software-only VMMs

- Binary translation
- Paravirtualization
- Device Emulation

Simpler and more Secure VMMs through foundation of virtualizable ISAs

Improved CPU and I/O virtualization **Performance and Functionality** as VMMs exploit infrastructure provided by VT-x, VT-i, VT-d

Past
No Hardware
Support

Today →
VMM software evolution over
time with hardware support



Intel Virtualization Technology

Intel® Virtualization Technology (Intel® VT)

Intel VT refers to all the hardware assists for virtualization that Intel offers across its platform (CPU, Chipset, I/O)

Intel VT for IA-32 and Intel® 64

(Intel VT-x, VT-x2)

Intel VT-x refers to all the hardware assists for virtualization in Intel® Xeon® processors



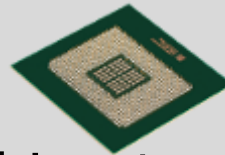
Processor

- Interrupt acceleration
- Flexibility in live migration
- Extended page tables

Intel VT For Directed I/O

(Intel VT-d, VT-d2)

Intel VT-d refers to all the hardware assists for virtualization in Intel chipset



Chipset

- I/O performance through direct assignment
- Memory protection

Intel VT For Connectivity

(Intel VT-c)

Intel VT-c refers to all the hardware assists for virtualization in Intel networking and I/O devices

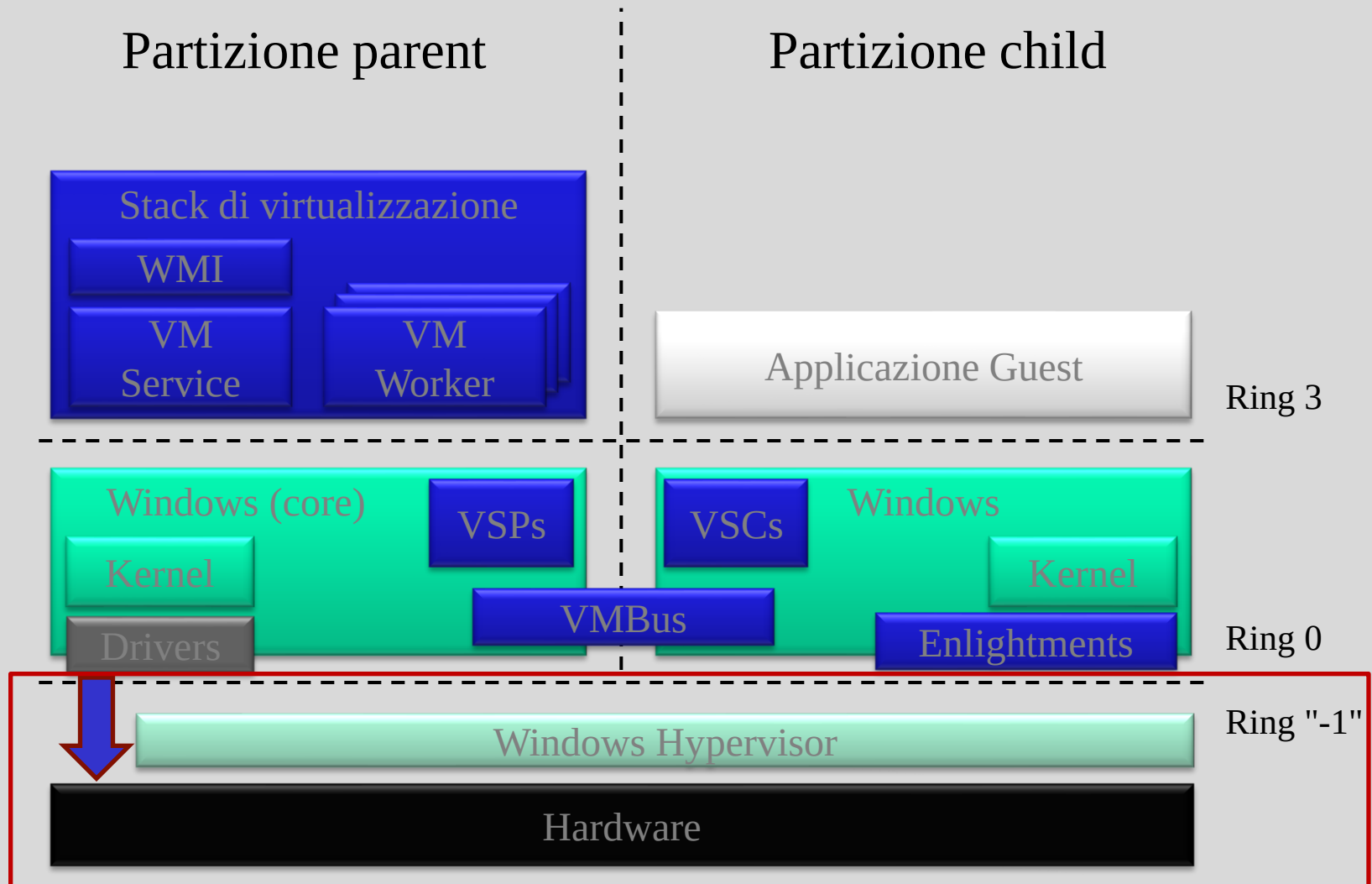


Network

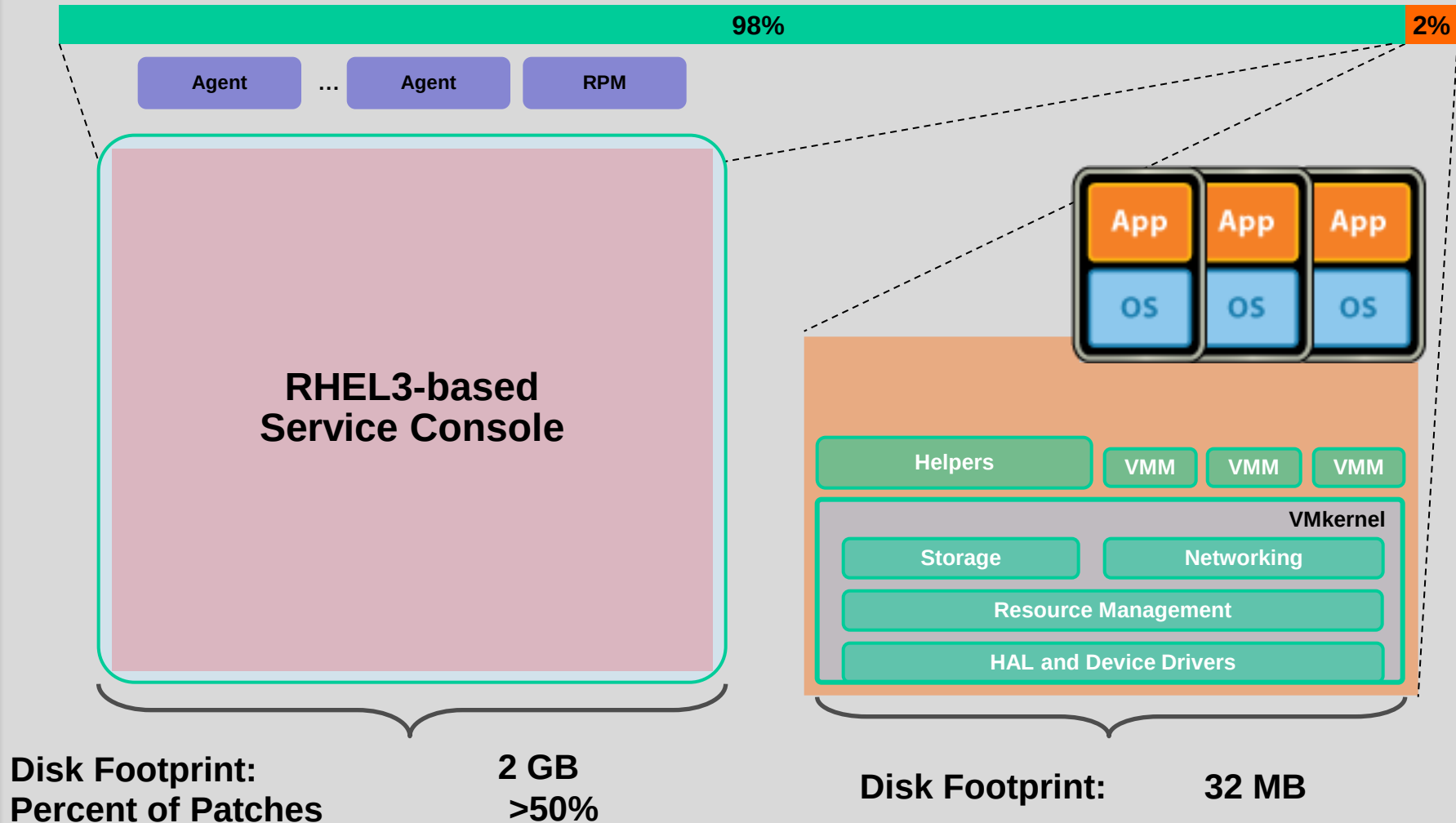
- Over 2x throughput gains on 10GbE
- Offload routine I/O tasks

<http://www.intel.com/technology/computing/vptech/>

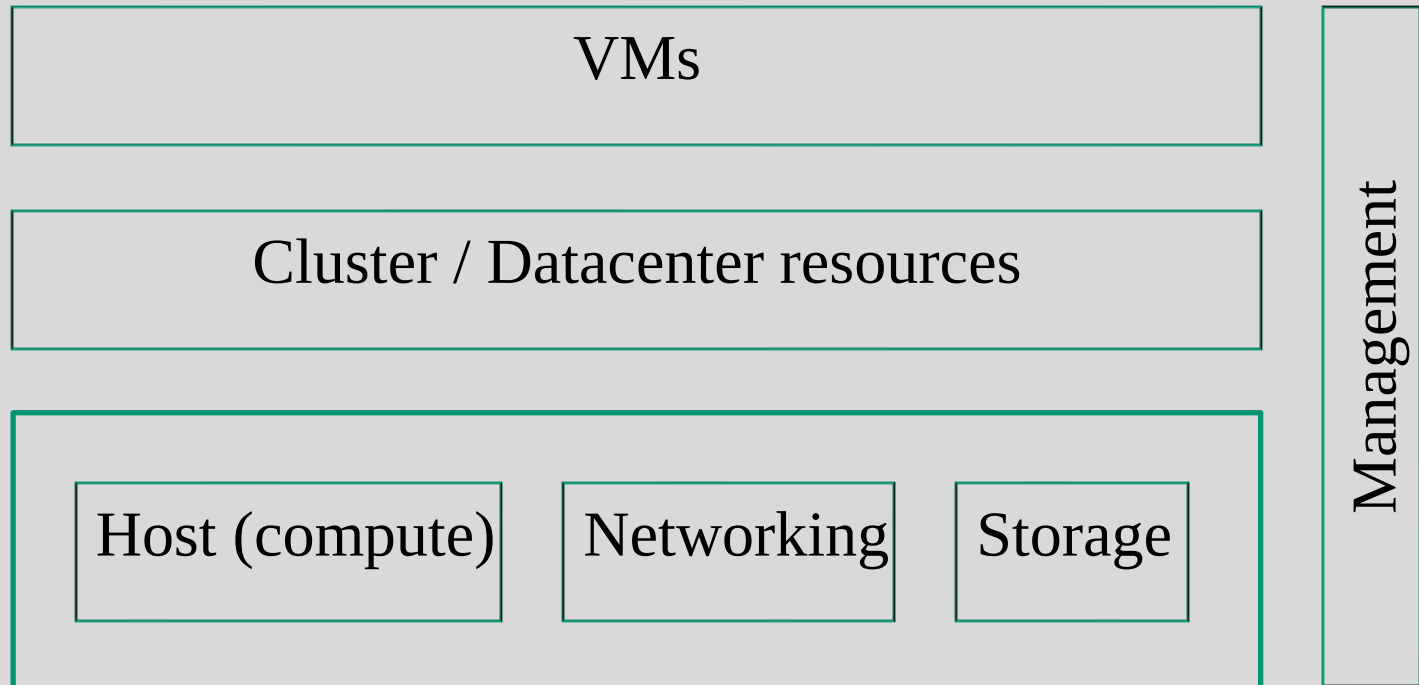
Microsoft Hyper-V



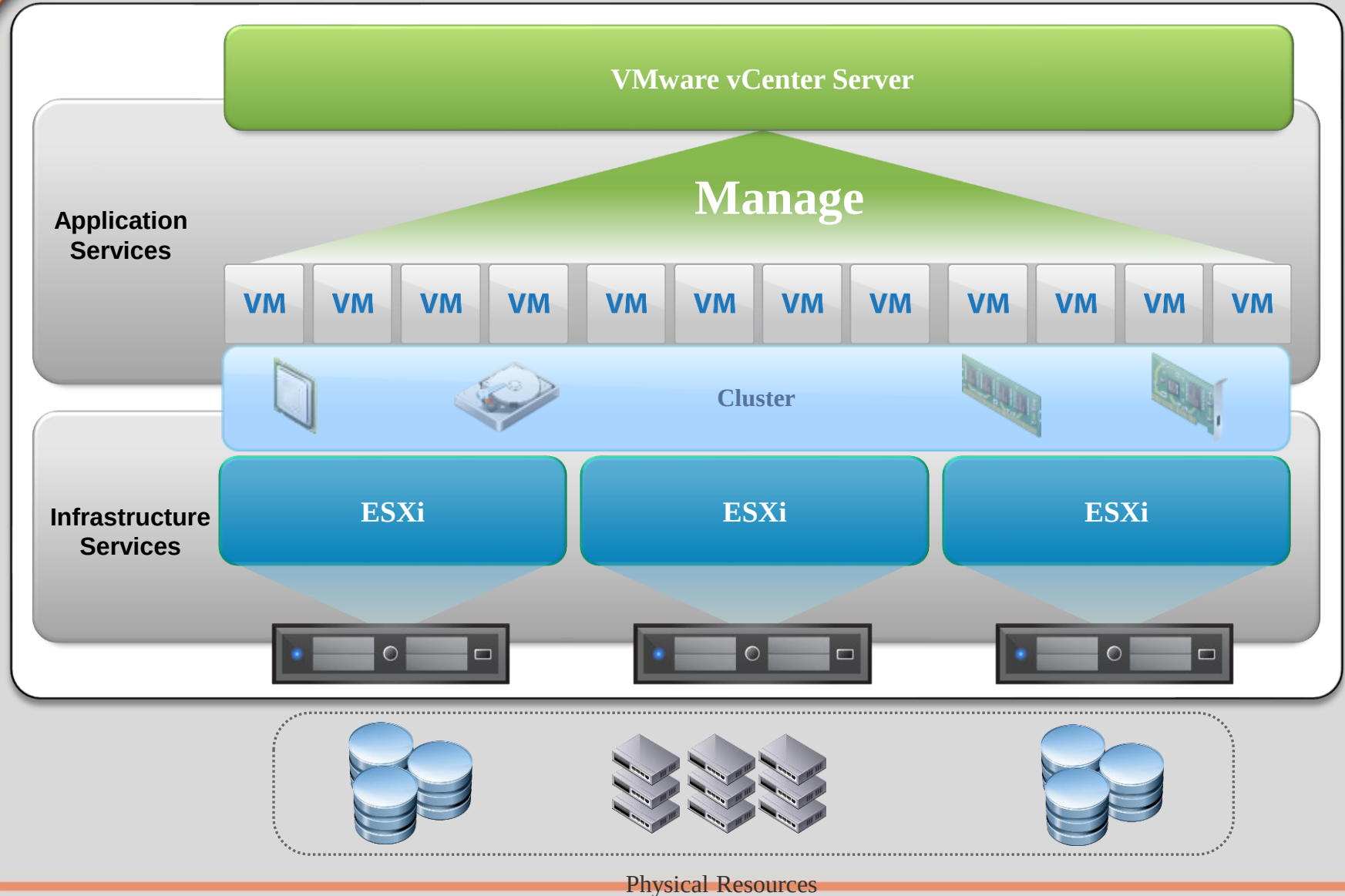
VMware ESX Server



Virtualizzazione enterprise



VMware vSphere



Gestione della virtualizzazione

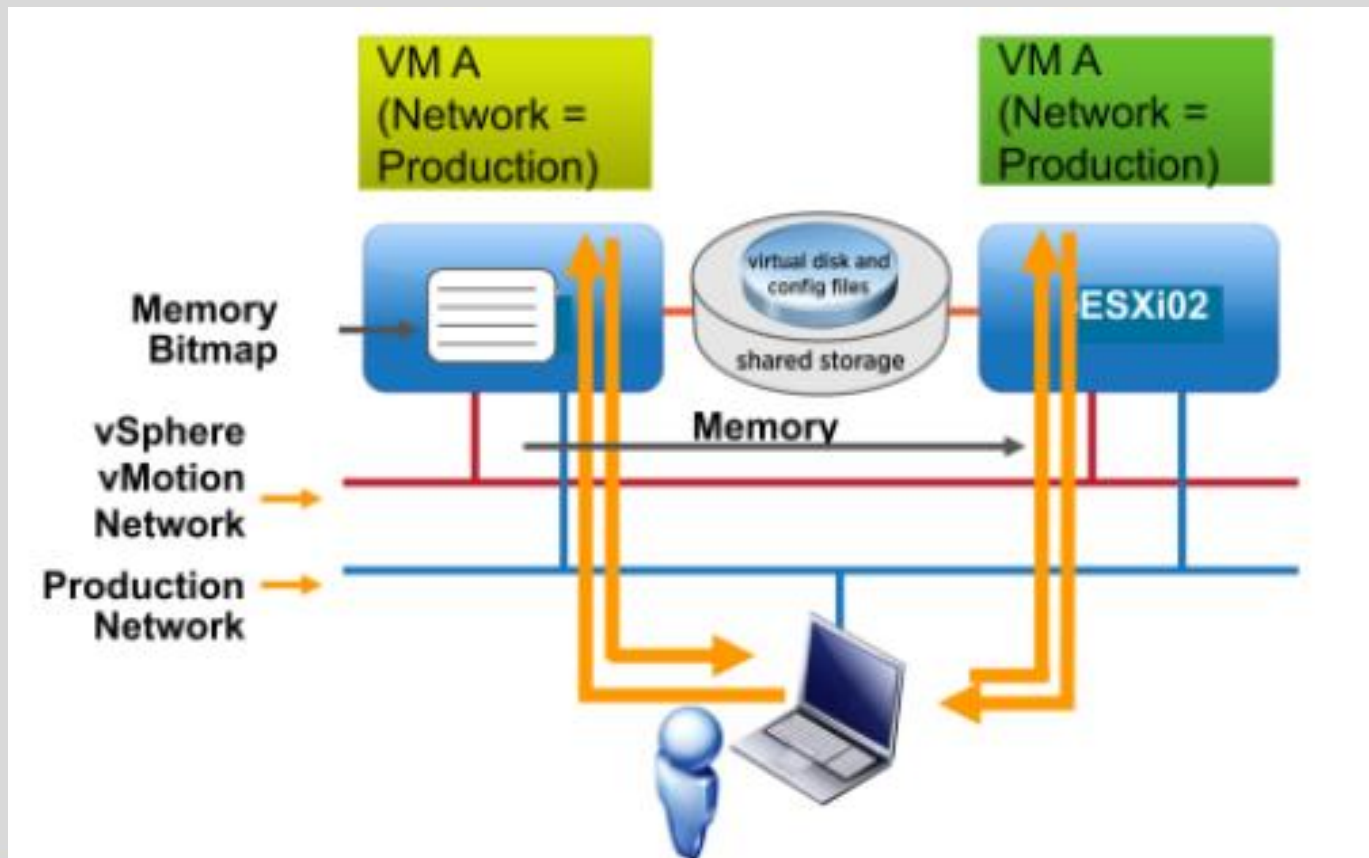
- VMware Virtual Center
 - Gestione centralizzata per host e VM
 - Diversi servizi
 - Cluster
- Xen Enterprise
- Virtual Iron
- Microsoft SystemCenter Virtual Machine Manager

VM migration

- Diverse modalità
 - Cold migration
 - Live migration
- Different types of migration
 - Migration across hosts
 - Migration across datastores
- Migration across cloud/hypervisor
 - Problema del «formato» delle VM
 - Problema del «formato» dei dischi virtuali

Live VM migration across hosts

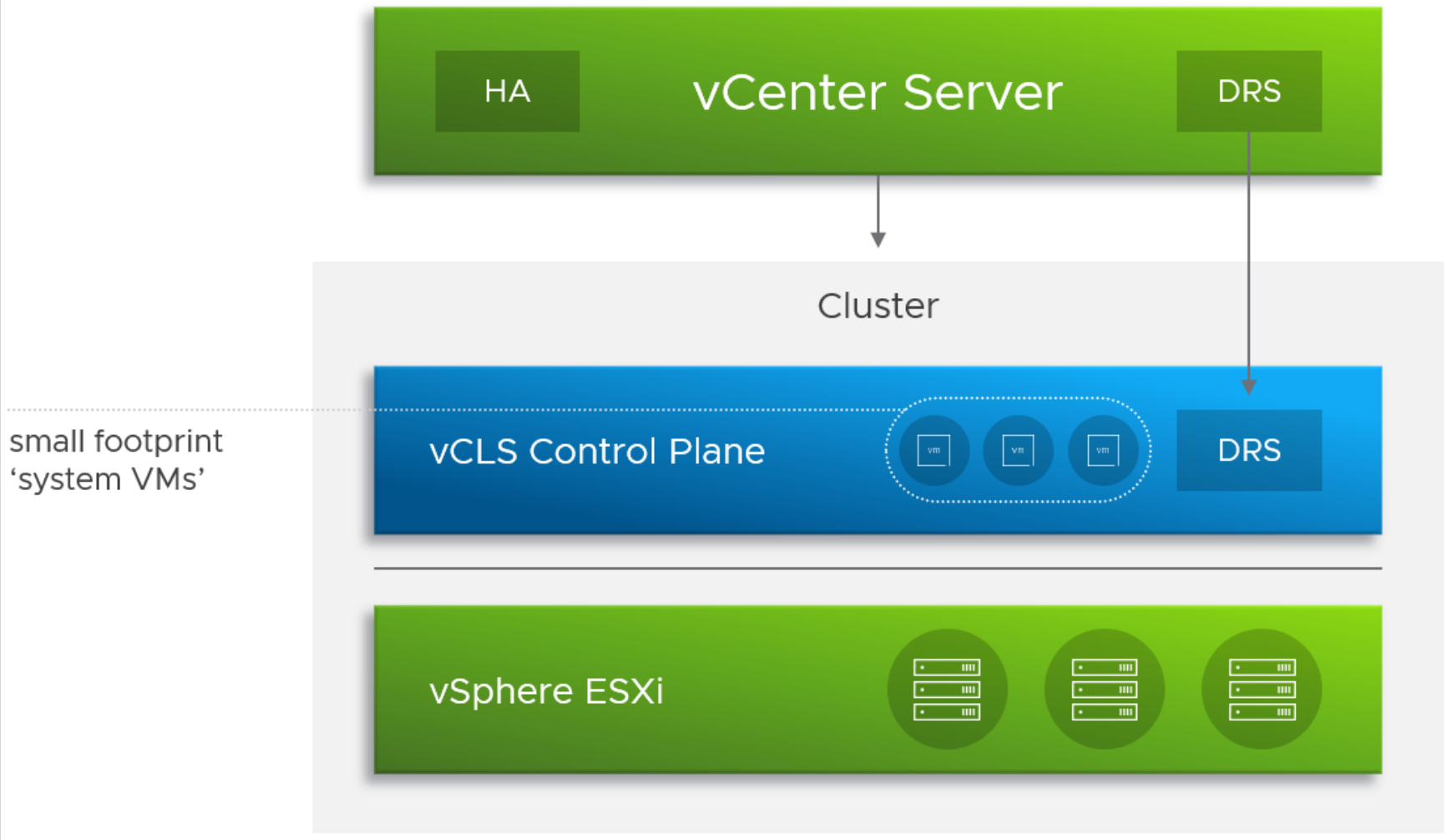
- Con o senza store condiviso
- Funzione introdotta nel 2003 da VMware



Altri servizi

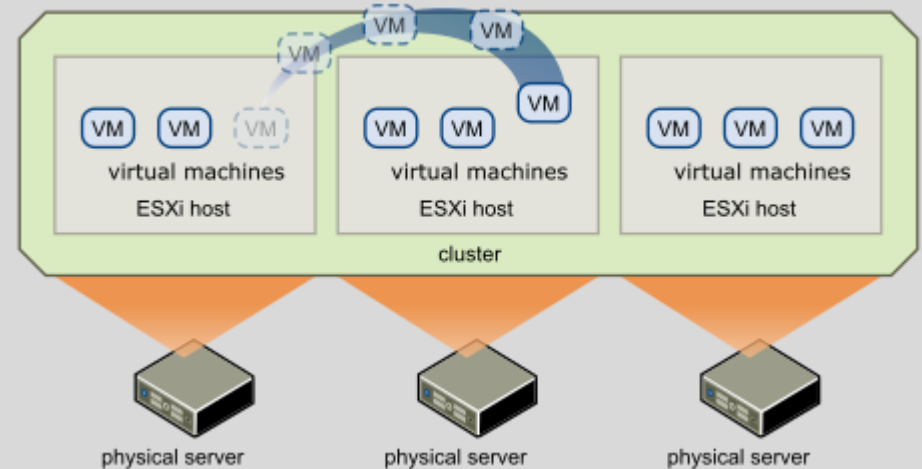
- VM Snapshot
- VM Cloning
- VM hot-add/remove
- VM hot-backup
- Template
- Monitoring
- Alerting
- ...

Control/Management/Data Plane



Cluster di virtualizzazione

- Gruppo di host con hypervisor type 1
- Le VM sono i servizi
- Funzionalità offerte dal cluster
 - VM migration
 - HA
 - LB
 - FT

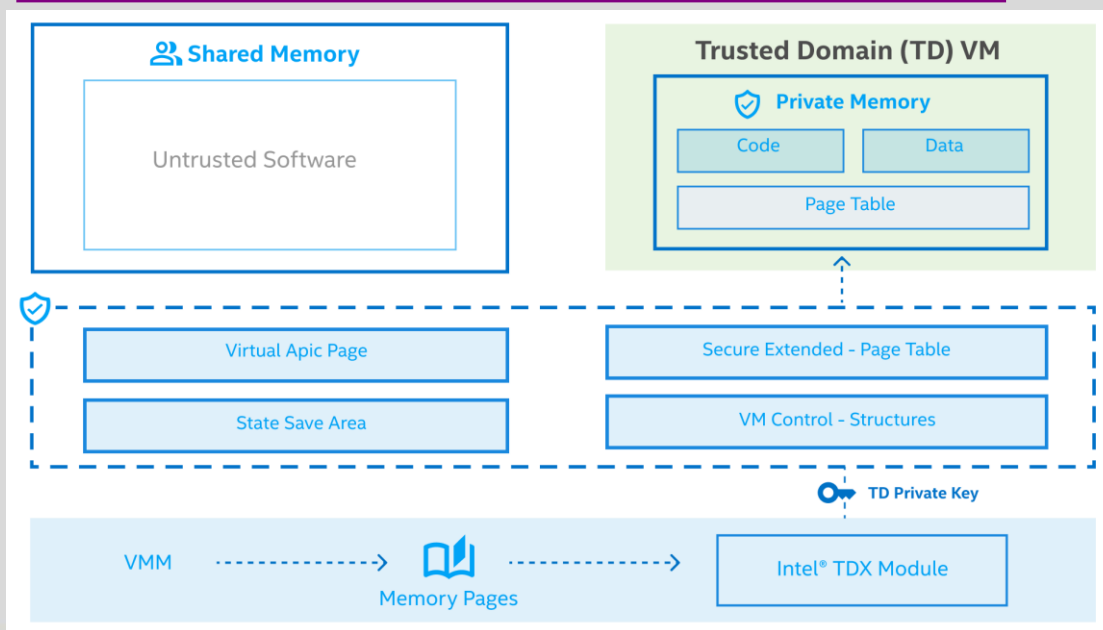


Virtualizzazione e sicurezza

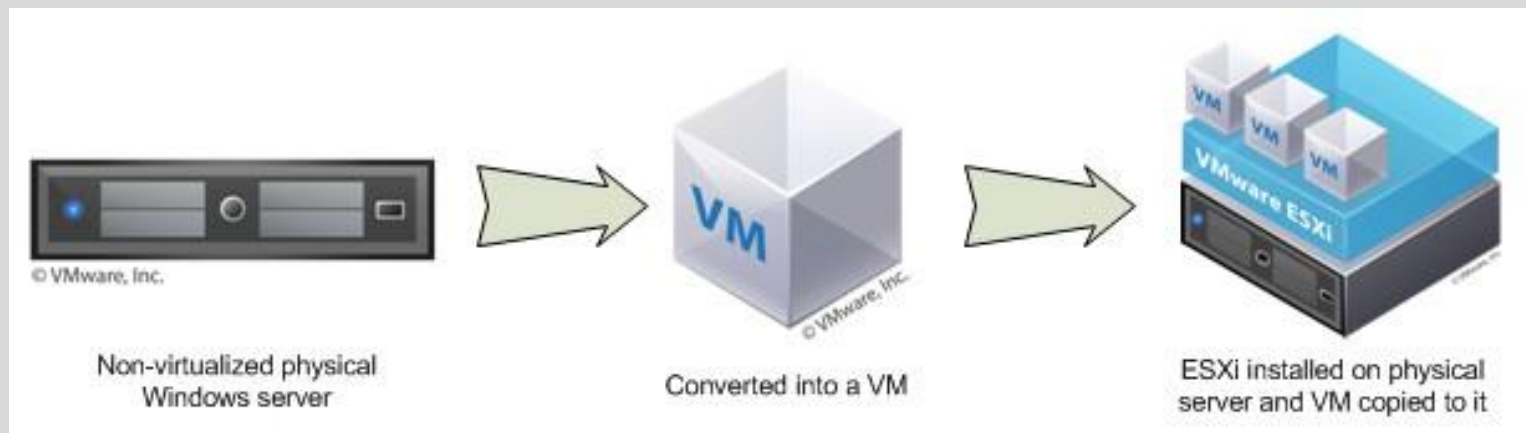
- Proteggere l'hypervisor dalle VM
 - Principio di isolamento?
 - Bug dell'hypervisor
 - Bug dell'hardware
- Proteggere le VM dall'hypervisor o dagli amministratori della virtualizzazione
 - VM Trusted Platform
- Proteggere le comunicazioni di rete
 - Live VM Migration
 - Storage condivisi

Sicurezza in hardware

- TPM e vTPM
- Intel Software Guard Extensions (SGX)
- Intel® Trust Domain Extensions (Intel® TDX)
 - <https://www.intel.com/content/www/us/en/developer/tools/trusted-domain-extensions/overview.html>



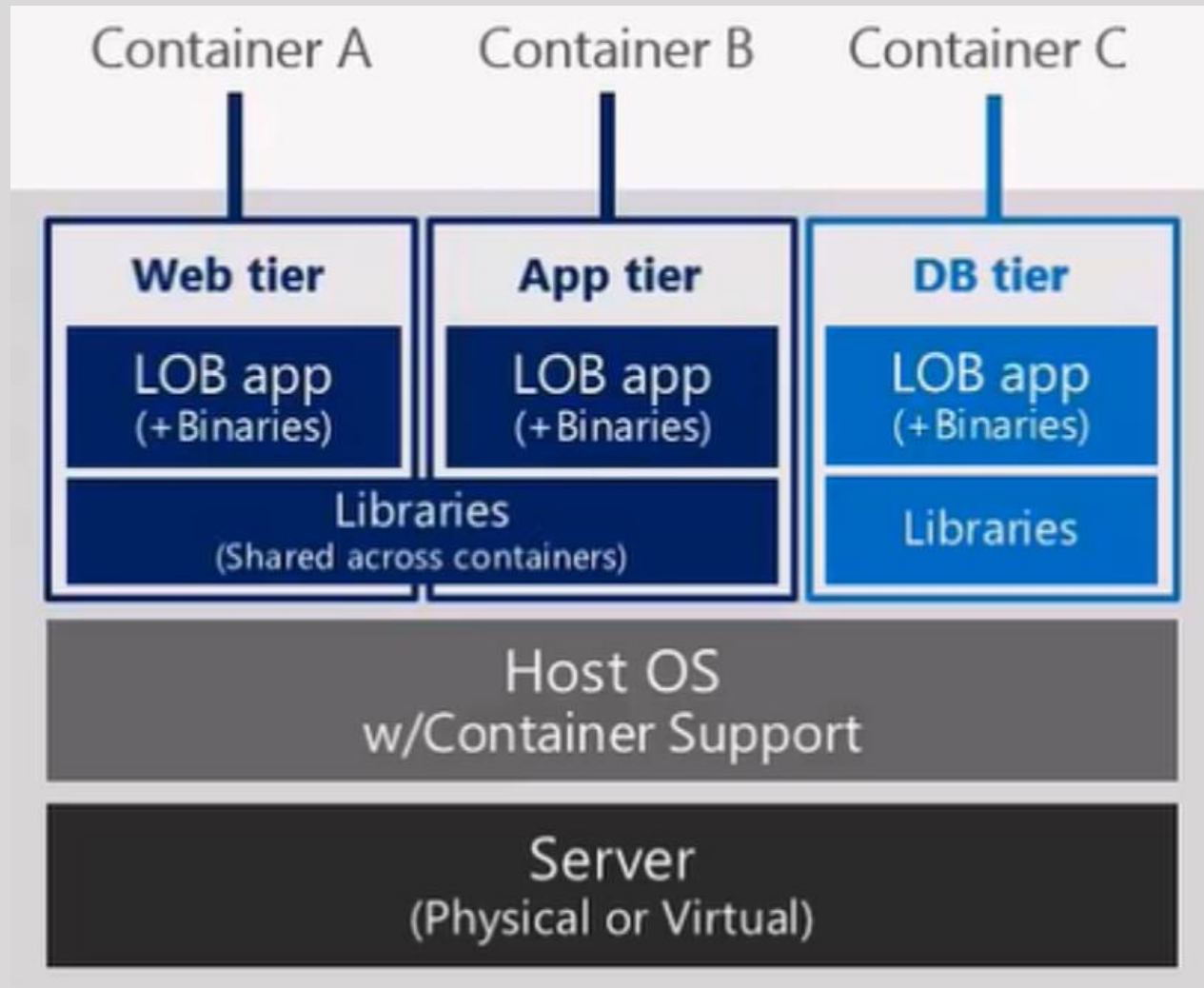
Da server a VM



Virtualizzazione

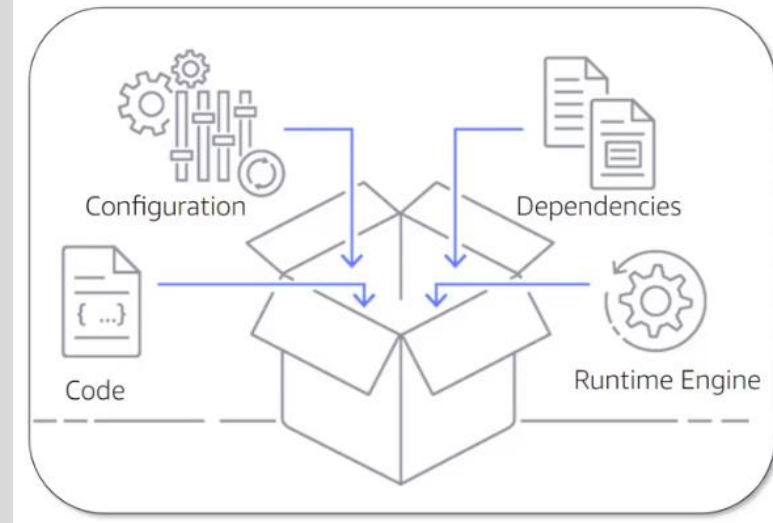
Container

Da server a container



Containers

- Logical object
- Standard packaging
- Portable
- Isolation
- Efficiency
- Fast
- Stateless or Statefull

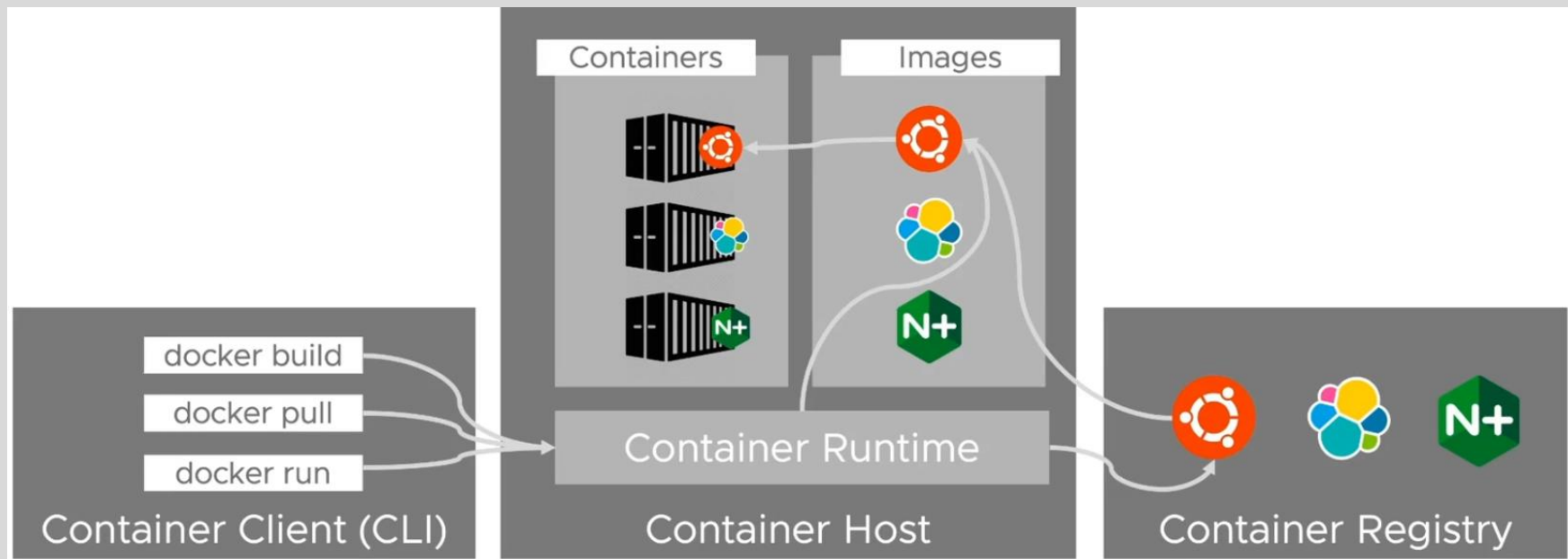


Containers are an application-centric method to deliver high-performing, scalable applications on any infrastructure of your choice.

Containers

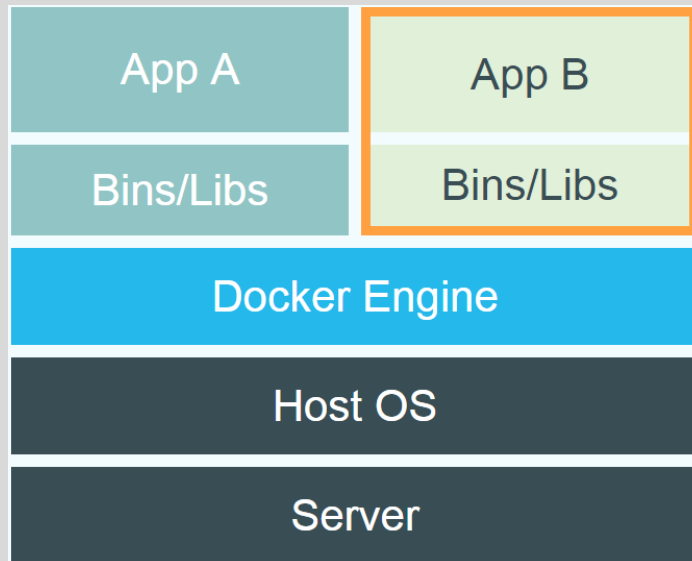
- Container
 - Tecnologia di virtualizzazione parziale che opera a livello di sistema operativo
- Evoluzione di soluzioni già esistenti
 - OpenVZ e Linux-Vserver
 - jail di FreeBSD, workload partitions AIX e i container/zone di Solaris
 - Linux namespaces e cgroups
- In forte crescita nel mondo cloud
 - Applicazioni Cloud Native
- Sistemi operativi per i containers
 - LXC (Linux Containers): dal kernel 2.6.24
 - Distribuzioni: CoreOS, PhotonOS, ...
 - *Process extracted from tarballs anchored to namespaces and controlled by cgroups*
 - Windows Containers: da Windows Server 2016

Terminologia

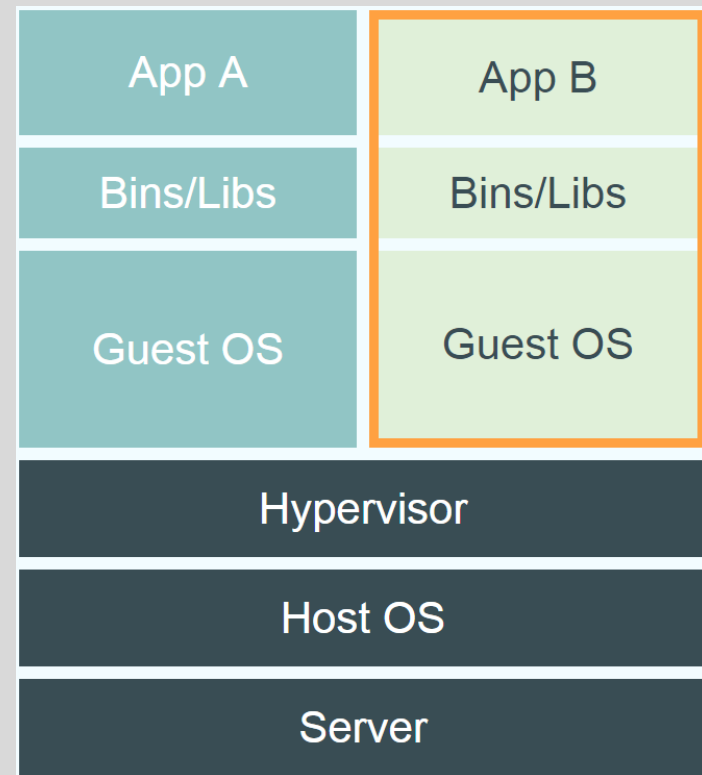


Containers vs. Virtualizzazione

- Container



- VM



Virtual Machines vs. Containers

Virtual Machines

- Full OS on virtual hardware
- Deployed as a unit
- Require a hypervisor
- VM performance critical to cloud performance
- Overhead cannot be removed
- Hotplugging requires support from guest OS

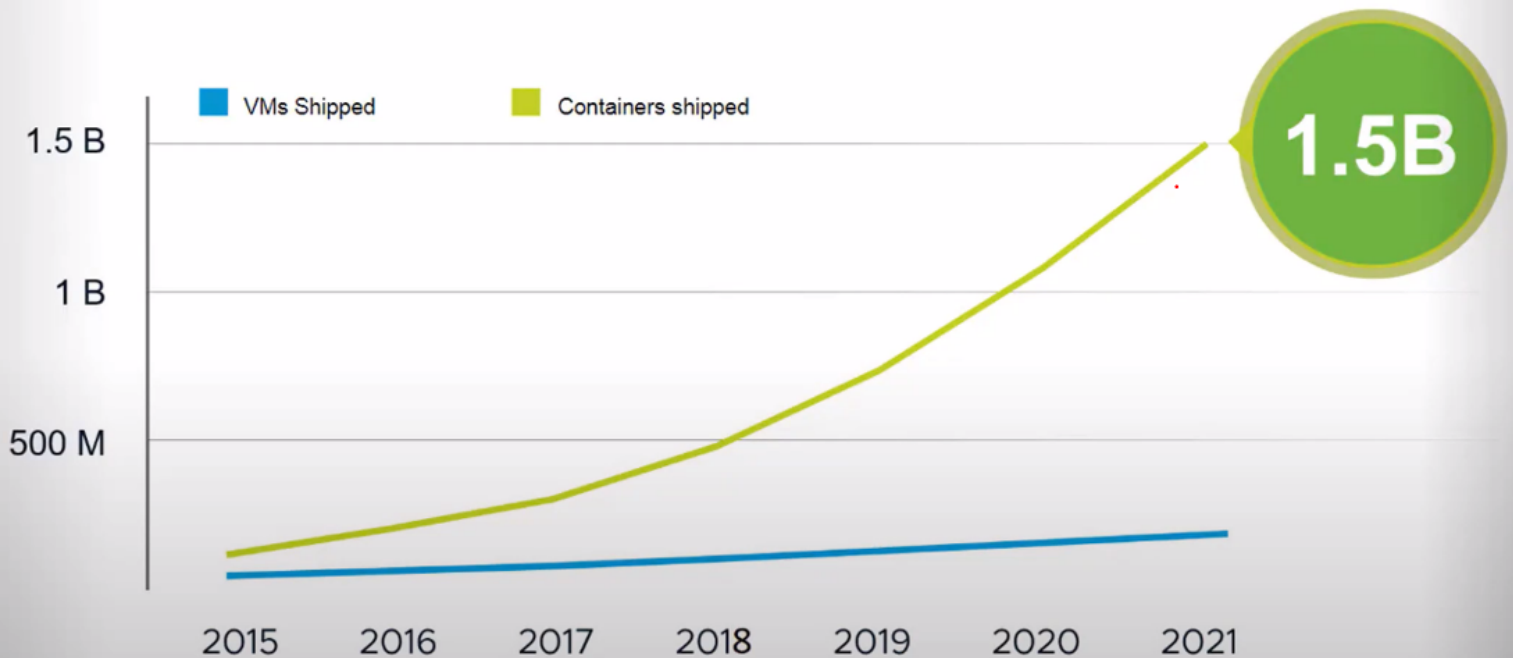
Containers

- Decouple application and OS
- Can be composed together
- Modifies existing OS to provide isolation
- A concept built on kernel namespace feature
- Contain as little as one process
- An application container consumes less RAM
- Operate at a higher abstraction level, offer more insight into behaviors without deploying additional agents

VMs vs. Containers

Benefits of Containers

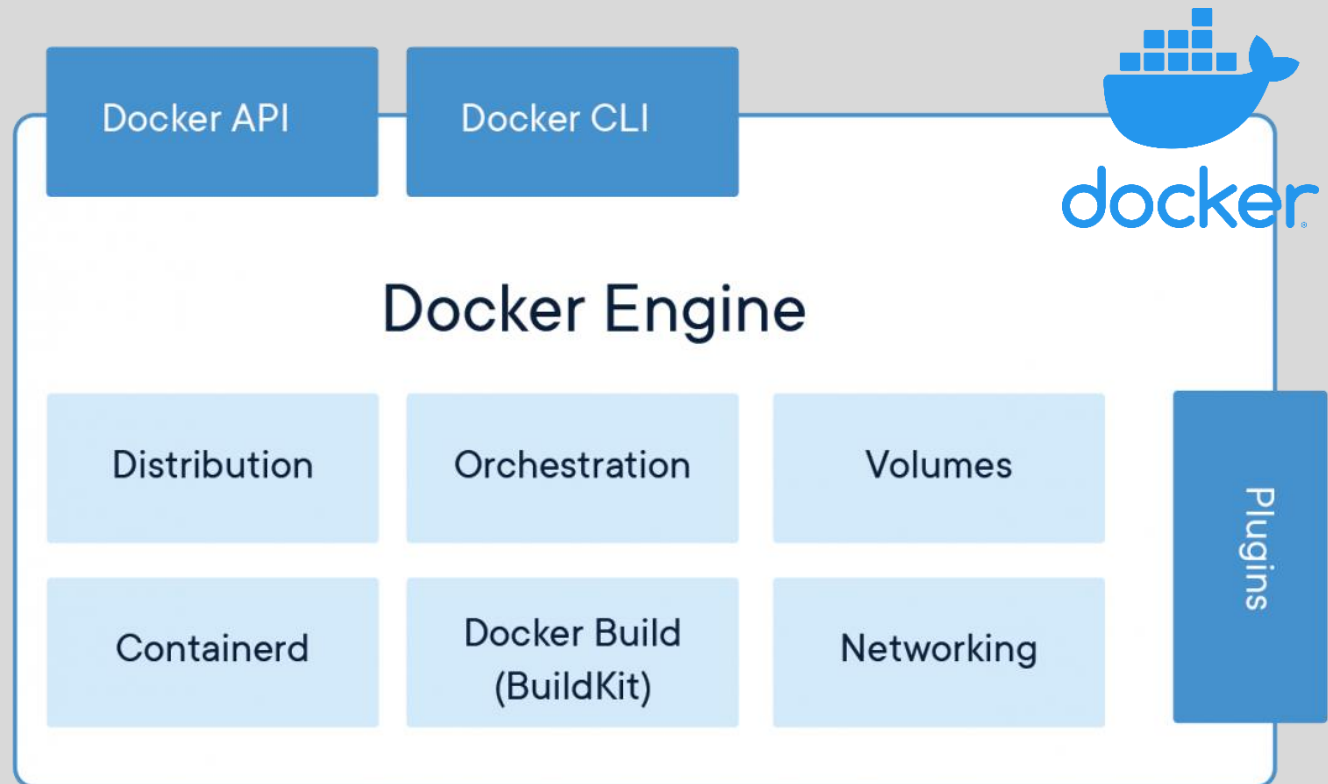
Containers are becoming the preferred deployment option for software.



Source: IDC, Matt Eastwood, IDC Directions 2019: Market Update, February 2019

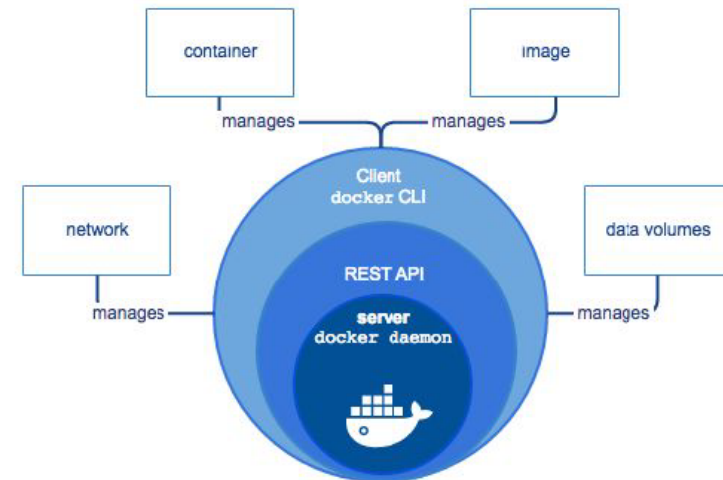
Container runtime

- Standard de fact
 - Docker Engine
 - open-source project that automates the deployment of applications inside software containers, by providing an additional layer of abstraction and automation of operating system-level virtualization on Linux
- Altre soluzioni
 - Containerd
 - CRI-O
 - runC



The Docker Engine

- The Docker Daemon/Engine
 - Manages everything
 - Exposes a REST API
- The Docker CLI
 - Interacts with the API
 - Provides a CLI-based user interface
 - Can be configured to point to remote engines

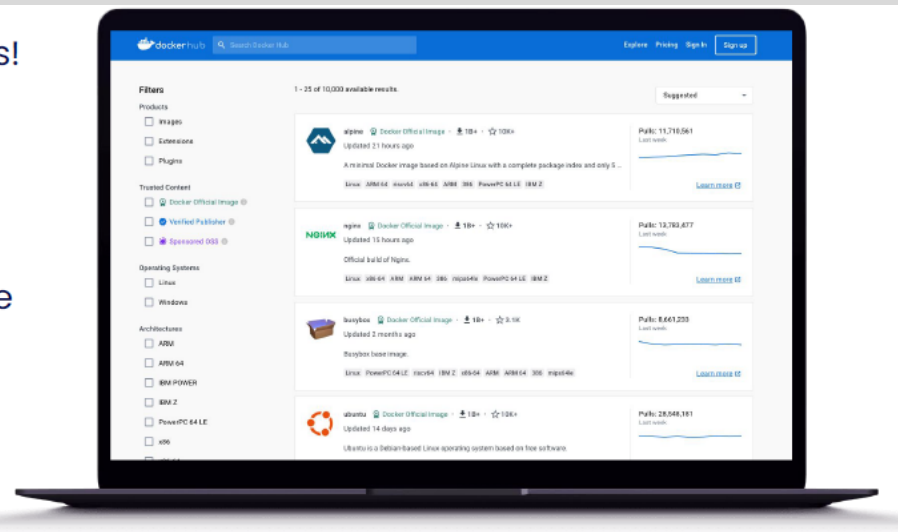


Container registry

- A container registry provides a central location for container image storage
- Types of registries
 - Hosted: Docker Hub, Google Container Registry
 - Self-Hosted: Artifactory, Harbor, Quay
- Images are built on a build host and pushed to a registry

Docker Hub

- The default marketplace for images!
- Trusted Content
 - Docker Official Images
 - Verified Publishers
 - Sponsored Open Source Software
- Personal and organizational repos



Docker da CLI

```
# Start a mongo container, named mongo, in the background
```

```
docker run --name mongo -d mongo
```

```
# Start an interactive ubuntu container and auto-remove it when it exits
```

```
docker run --rm -it ubuntu
```

```
# Start nginx and expose its web server port to the host
```

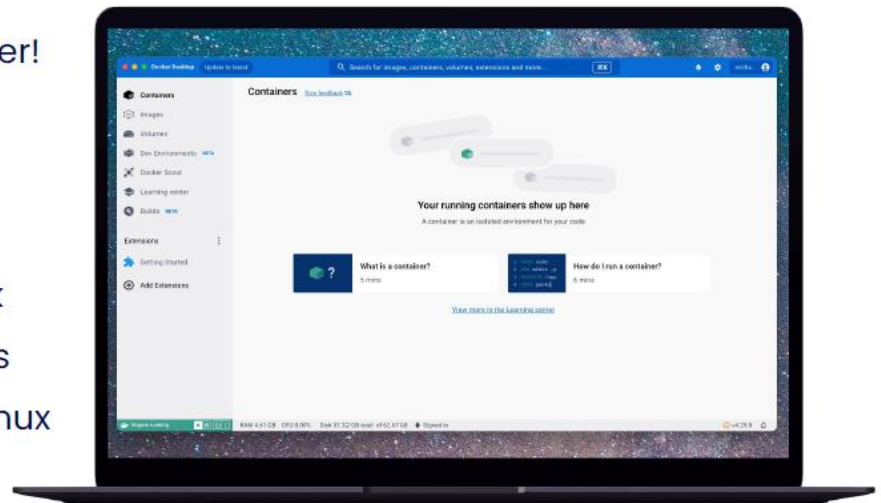
```
# The port mapping is [host port]:[container port]
```

```
docker run -d -p 80:80 nginx
```

Docker da GUI

Docker Desktop

- The local install for everything Docker!
- Provides the ability to...
 - Run containerized workloads
 - Build images
 - Launch Kubernetes with one click
- Provides both CLI and GUI interfaces
- Works across Mac, Windows, and Linux



Risorse di un container

- Risorse dell'host
- Disco in overlay
- Rete in virtual switch su network privata

```
root@45df068a2a45:/# df -h
Filesystem      Size  Used Avail Use% Mounted on
overlay         1007G  1.5G  955G   1% /
tmpfs           64M    0   64M   0% /dev
tmpfs           5.8G    0   5.8G   0% /sys/fs/cgroup
shm            64M    0   64M   0% /dev/shm
/dev/sdd        1007G  1.5G  955G   1% /etc/hosts
tmpfs           5.8G    0   5.8G   0% /proc/acpi
tmpfs           5.8G    0   5.8G   0% /sys/firmware
```

```
root@45df068a2a45:/# free
              total        used        free      shared  buff/cache   available
Mem:      12097464    1260452     8539908         6036     2297104    10554776
Swap:      3145728           0     3145728
root@45df068a2a45:/#
```

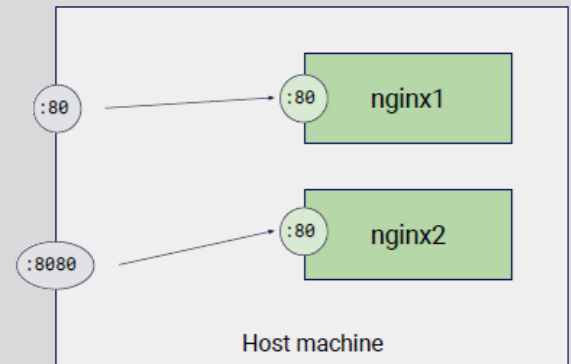
```
scsi 0:0:0:0: Direct-Access    Msft      Virtual Disk        1.0  PQ: 0 ANSI: 5
sd 0:0:0:0: Attached scsi generic sg0 type 0
sd 0:0:0:0: [sda] 744080 512-byte logical blocks: (381 MB/363 MiB)
sd 0:0:0:0: [sda] Write Protect is on
sd 0:0:0:0: [sda] Mode Sense: 0f 00 80 00
sd 0:0:0:0: [sda] Write cache: disabled, read cache: enabled, doesn't support DPO or FUA
EXT4-fs (sda): mounted filesystem without journal. Opts: (null). Quota mode: none.
sd 0:0:0:0: [sda] Attached SCSI disk
scsi 0:0:0:1: Direct-Access    Msft      Virtual Disk        1.0  PQ: 0 ANSI: 5
sd 0:0:0:1: Attached scsi generic sg1 type 0
sd 0:0:0:1: [sdb] 6291464 512-byte logical blocks: (3.22 GB/3.00 GiB)
sd 0:0:0:1: [sdb] 4096-byte physical blocks
sd 0:0:0:1: [sdb] Write Protect is off
sd 0:0:0:1: [sdb] Mode Sense: 0f 00 00 00
sd 0:0:0:1: [sdb] Write cache: enabled, read cache: enabled, doesn't support DPO or FUA
sd 0:0:0:1: [sdb] Attached SCSI disk
Adding 3145728k swap on /dev/sdb. Priority:-2 extents:1 across:3145728k
```


Container e rete

Explaining port mappings

- Every container gets its own IP address
- When a container uses a port, it's only listening on its IP address
- Exposing a port = create config to connect the host network interfaces to the container

```
docker run -p 80:80 --name nginx1 nginx  
docker run -p 8080:80 --name nginx2 nginx
```

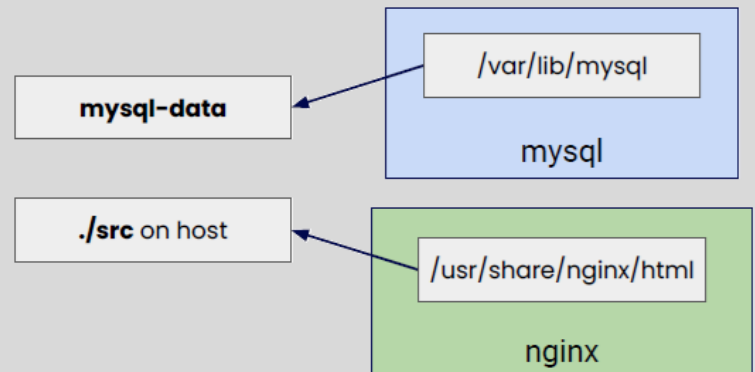


Container e storage

- The container's filesystem comes from the image
- Bind mounts let us share files from the host
- Volumes let us persist data in the container

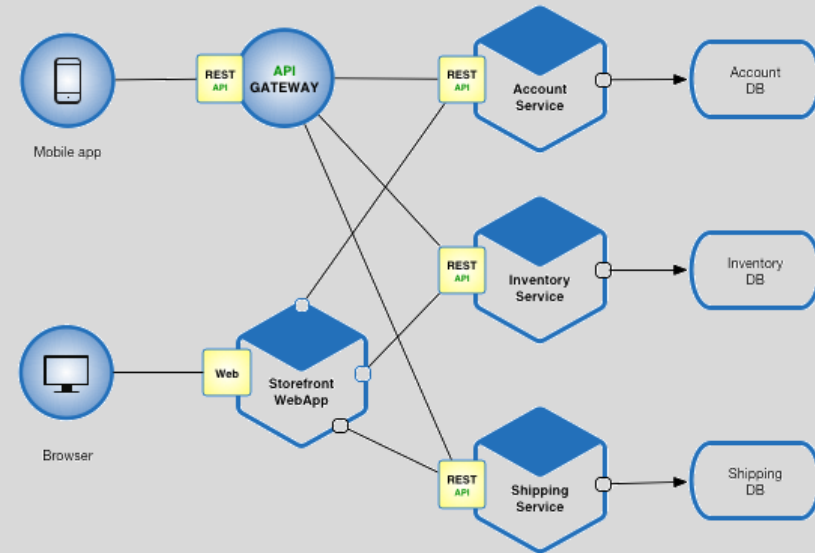
```
docker run -v mysql-data:/var/lib/mysql ... mysql
```

```
docker run -v ./src:/usr/share/nginx/html ... nginx
```

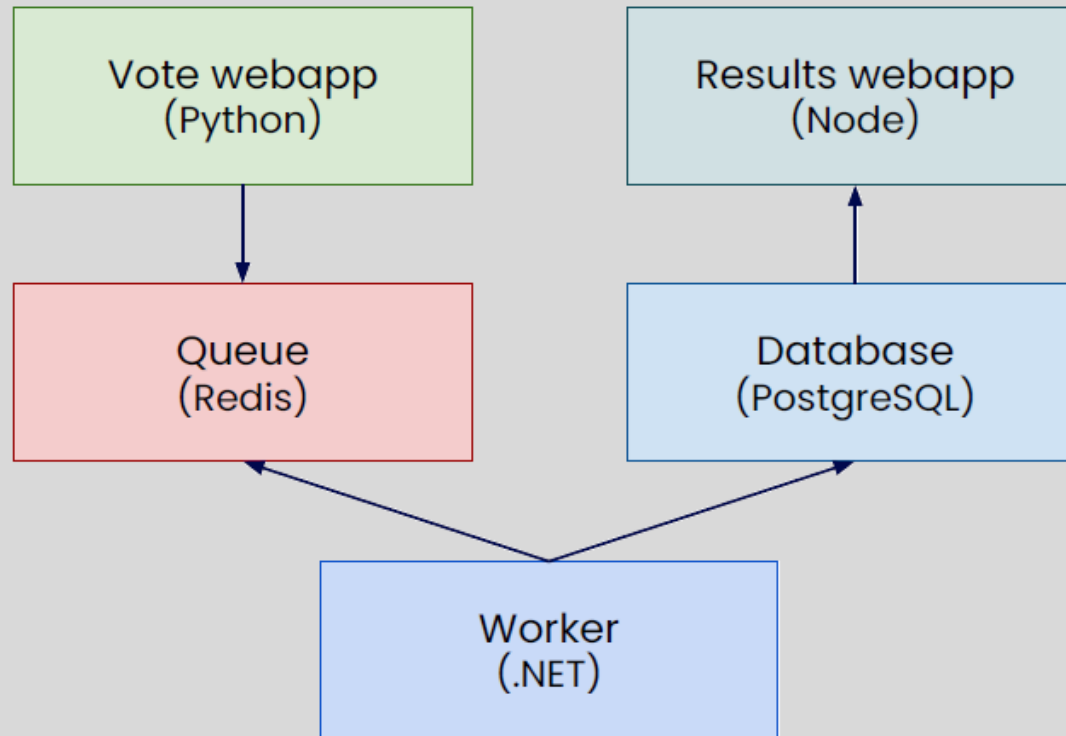


Microservices

- <https://microservices.io/patterns/microservices.html>
- <https://microservices.io/>
- Also known as the microservice architecture
Architectural style that structures an application as a collection of services that are
 - Highly maintainable and testable
 - Loosely coupled
 - Independently deployable
 - Organized around business capabilities
 - Owned by a small team
- The microservice architecture enables the rapid, frequent and reliable delivery of large, complex applications
 - also enables an organization to evolve its technology stack



Applicazione con più micro-servizi



Example voting app: github.com/docker-samples/example-voting-app

Problematiche dei container

- Definizioni di standard
- Gestione delle risorse
 - Gestione dei container e relativo discovery dei servizi
 - Gestione delle immagini
 - Gestione del computing
- Cloud Native Networking
 - Gestione della rete
- Cloud Native Storage
 - Gestione dello storage e della persistenza
 - Volume driver
 - Local and remote storage management

Gestione dei containers

- Standard de facto
 - Google Kubernetes
 - Disponibile anche as a Service (AKS, EKS, GKE, ...)
 - <https://kubernetes.io/docs/tutorials/kubernetes-basics/tutorial-kube/>
- Altre soluzioni
 - Docker Swarn
 - Mesosphere DC/OS | Marathon



docker
SWARM



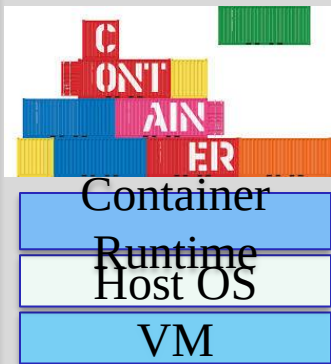
Apache
MESOS



kubernetes

From Containers to Kubernetes

Container



Benefits

Isolation
Immutable infrastructure
Portability
Faster deployments
Versioning
Ease of sharing

Challenges

Networking
Deployments
Service Discovery
Auto Scaling
Persisting Data
Logging,
Monitoring
Access Control



**Container
Scheduler**

Kubernetes

Orchestration of cluster of
containers across multiple hosts

- Automatic placements, networking, deployments, scaling, roll-out/-back, A/B testing

Declarative – not procedural

- Declare target state, reconcile to desired state
- Self-healing

Workload Portability

- Abstract from cloud provider specifics
- Multiple container runtimes



What is Kubernetes?

- **Kubernetes** is an open-source system for automating deployment, scaling, and management of (containerized) applications and services
- **Features:**
 - Run Anywhere
 - Automatic binpacking
 - Self-healing
 - Horizontal scaling
 - Service discovery and load balancing
 - Automated rollouts and rollbacks
 - Secret and configuration management
 - Storage orchestration



Funzionalità di Kubernetes

- Scoperta dei servizi e bilanciamento del carico
 - Kubernetes può esporre un container usando un nome DNS o il suo indirizzo IP
 - Se il traffico verso un container è alto, Kubernetes è in grado di distribuire il traffico su più container in modo che il servizio rimanga stabile
- Orchestrazione dello storage
 - Kubernetes ti permette di montare automaticamente un sistema di archiviazione di vostra scelta, come per esempio storage locale, dischi forniti da cloud pubblici, e altro ancora
- Rollout e rollback automatizzati
 - Puoi utilizzare Kubernetes per descrivere lo stato desiderato per i propri container, e Kubernetes si occuperà di cambiare lo stato attuale per raggiungere quello desiderato ad una velocità controllata
 - Per esempio, puoi automatizzare Kubernetes per creare nuovi container per il tuo servizio, rimuovere i container esistenti e adattare le loro risorse a quelle richieste dal nuovo container
- Ottimizzazione dei carichi
 - Fornisci a Kubernetes un cluster di nodi per eseguire i container
 - Puoi istruire Kubernetes su quanta CPU e memoria (RAM) ha bisogno ogni singolo container
 - Kubernetes allocherà i container sui nodi per massimizzare l'uso delle risorse a disposizione
- Self-healing
 - Kubernetes riavvia i container che si bloccano, sostituisce container, termina i container che non rispondono agli health checks, e evita di far arrivare traffico ai container che non sono ancora pronti per rispondere correttamente
- Gestione di informazioni sensibili e della configurazione
 - Kubernetes consente di memorizzare e gestire informazioni sensibili, come le password, i token OAuth e le chiavi SSH
 - Puoi distribuire e aggiornare le informazioni sensibili e la configurazione dell'applicazione senza dover ricostruire le immagini dei container e senza svelare le informazioni sensibili nella configurazione del tuo sistema

Modello imperativo vs. dichiarativo

Imperative

defines actions



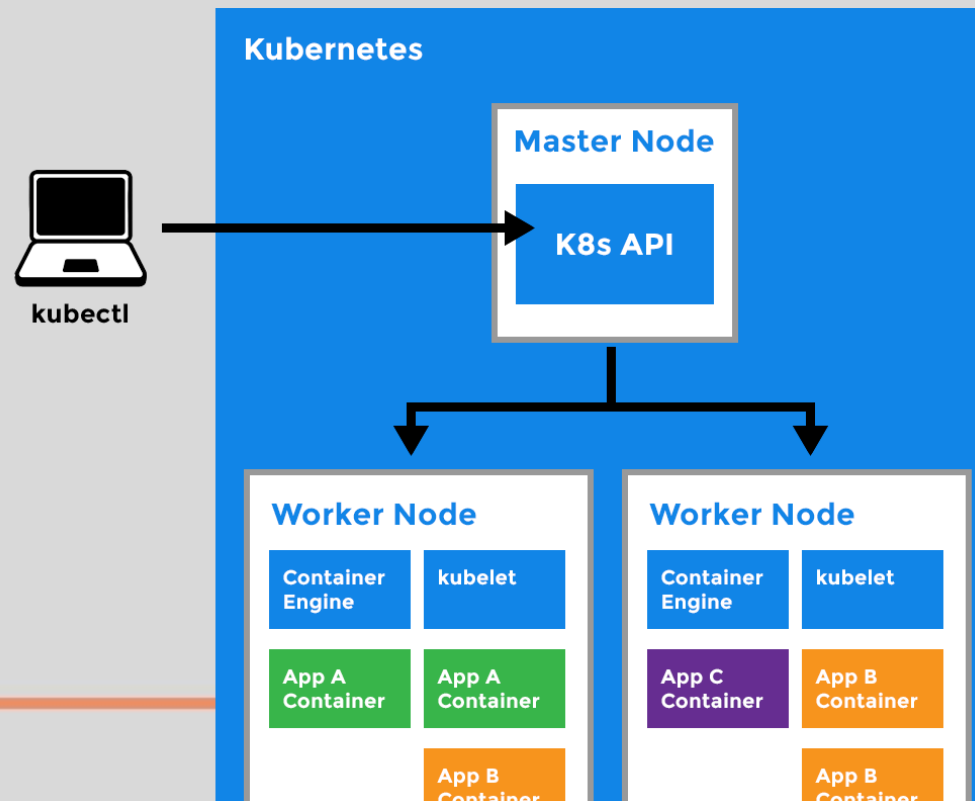
Declarative

defines desired state



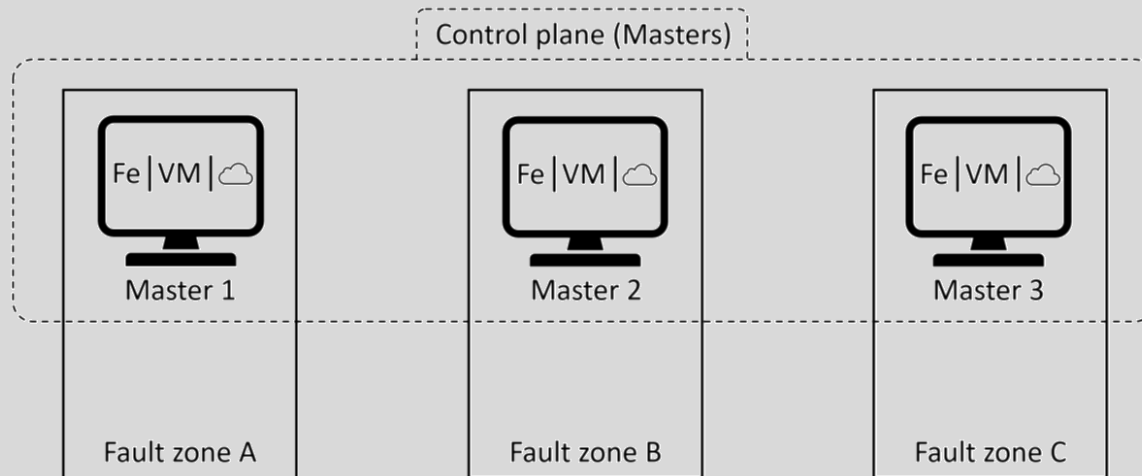
Kubernetes

- Kubernetes (commonly referred to as K8s)
 - orchestration engine for container technologies such as Docker and rkt
 - is taking over the DevOps scene in the last couple of years



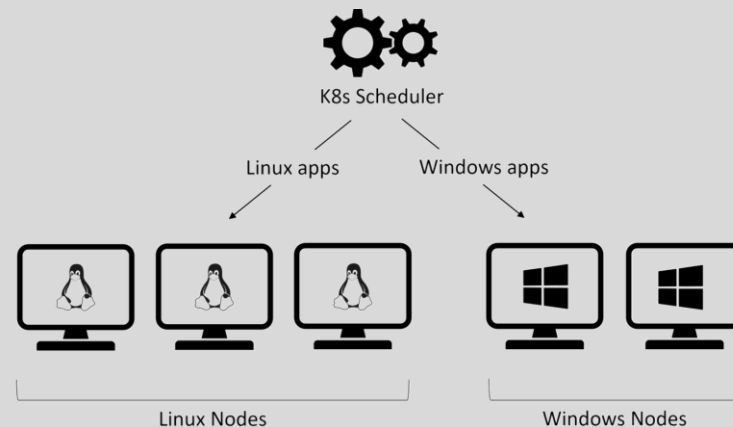
Kubernetes – Control Plane

- Kubectl
 - A CLI tool for Kubernetes
- Master Node
 - The main machine that controls the nodes
 - Main entrypoint for all administrative tasks
 - It handles the orchestration of the worker nodes



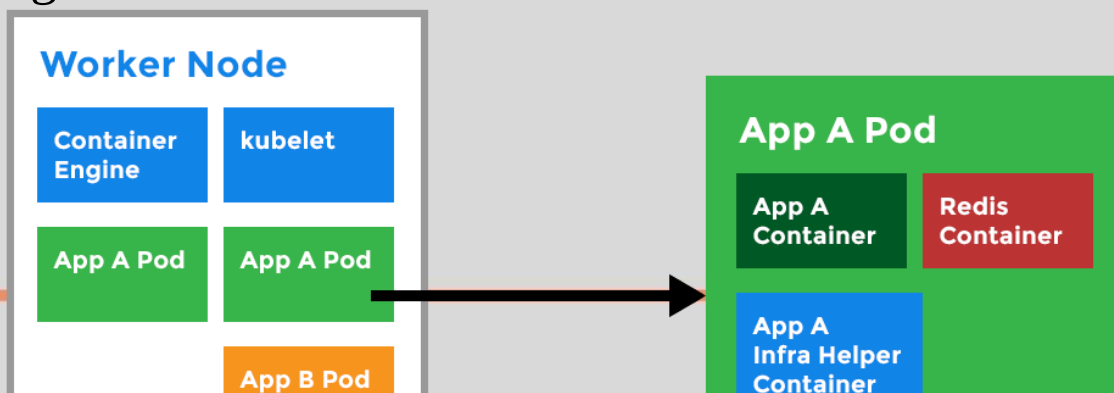
Kubernetes – Nodes

- Kubelet
 - Primary node agent
 - Ensures that containers are running and healthy
- Worker Node
 - It is a worker machine in Kubernetes (used to be known as minion)
 - This machine performs the requested tasks. Each Node is controlled by the Master Node
 - Runs pods (that have containers inside them)
 - This is where the Docker engine runs and takes care of downloading images and starting containers

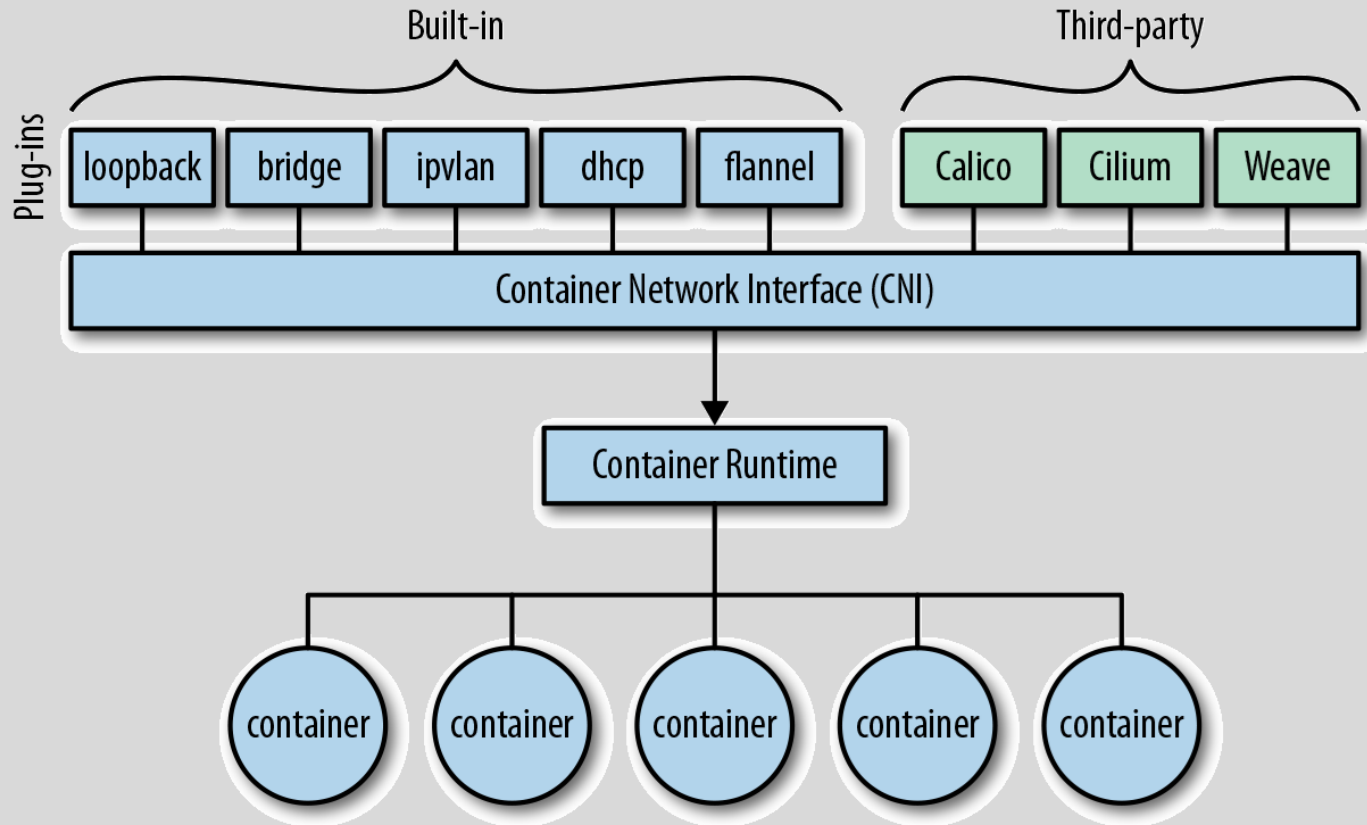


Kubernetes - Pod

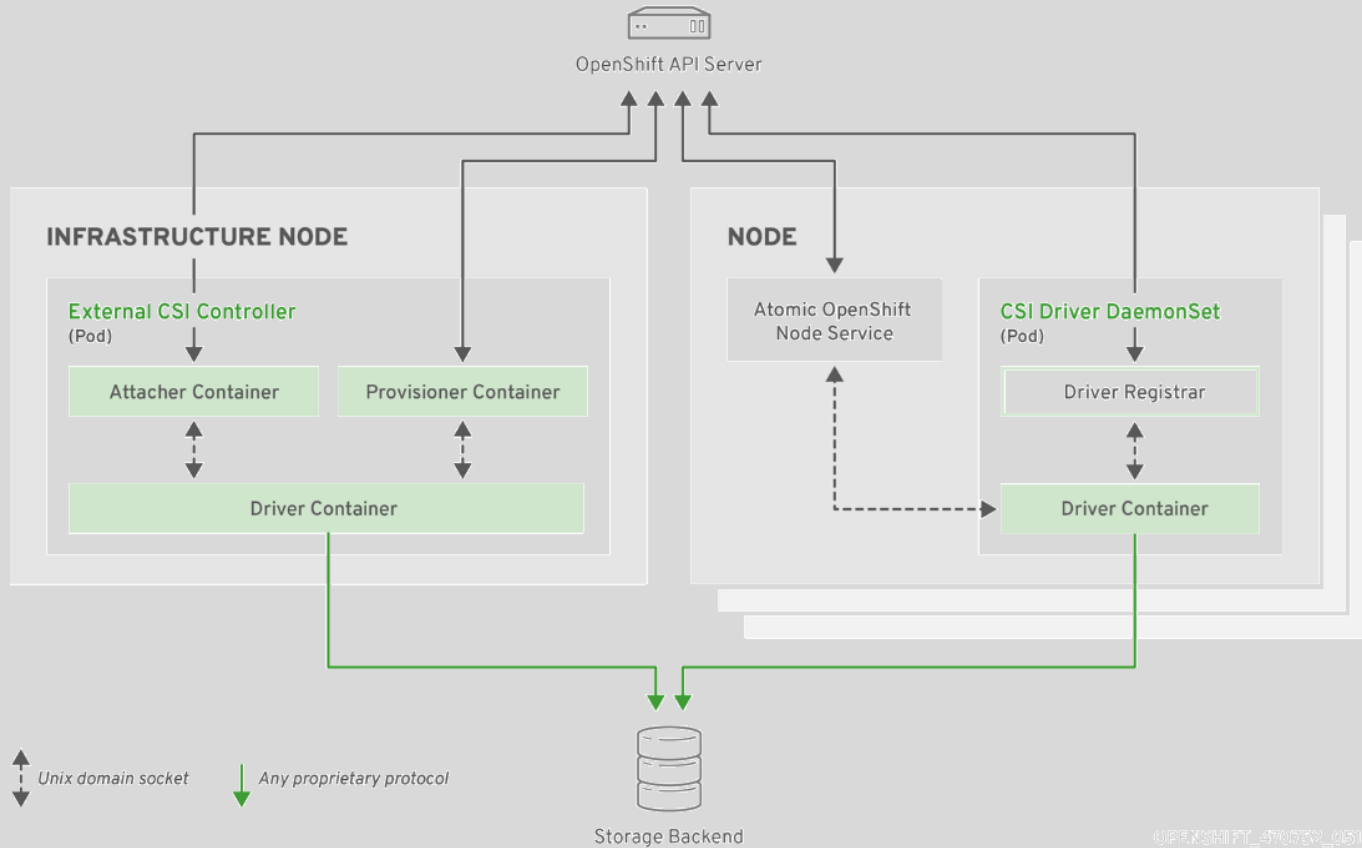
- Kubernetes Pod
 - A Pod can host multiple containers and storage volumes
 - Pods are instances of Deployments
 - One Deployment can have multiple pods
 - With Horizontal Pod Autoscaling, Pods of a Deployment can be automatically started and halted based on CPU usage
 - Containers within the same pod have access to shared volumes
 - Each Pod has its unique IP Address within the cluster
 - Pods are up and running until someone (or a controller) destroys them
 - Any data saved inside the Pod will disappear without a persistent storage



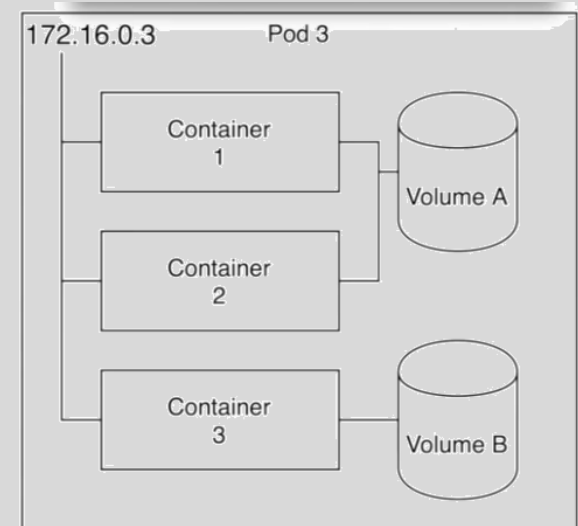
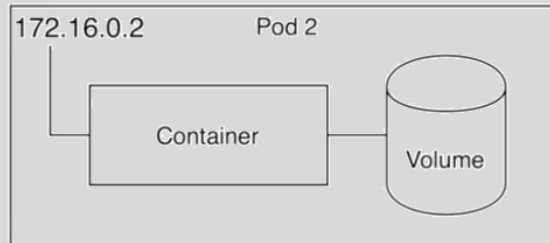
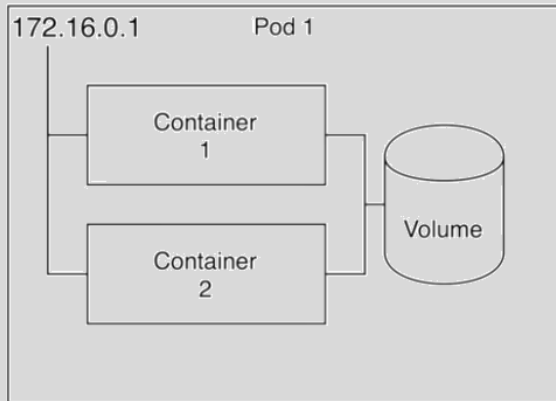
Container Network Interface (CNI)



Container Storage Interface (CSI)



Pod – Network & Storage



Cloud Native Readiness

Making applications successful atop Kubernetes

KUBERNETES AVAILABLE

Platform built atop Kubernetes made available for Developers.



DEVELOPER WORKFLOW

Application development, build, and testing lifecycles.



APPLICATION ENHANCEMENTS

Application practices to enable resilience, observability, and scaling of an application.



BUILD AND PACKAGING

Building, containerizing, versioning, and packaging applications.



APPLICATION LIFECYCLE

Automating advanced application lifecycles such as stateful workloads.



OBSERVABILITY

Tracing, logging, and monitoring practices for cloud native apps.



CLOUD NATIVE APPS

Application deploying, running, and scaling successfully on cloud native infrastructure.



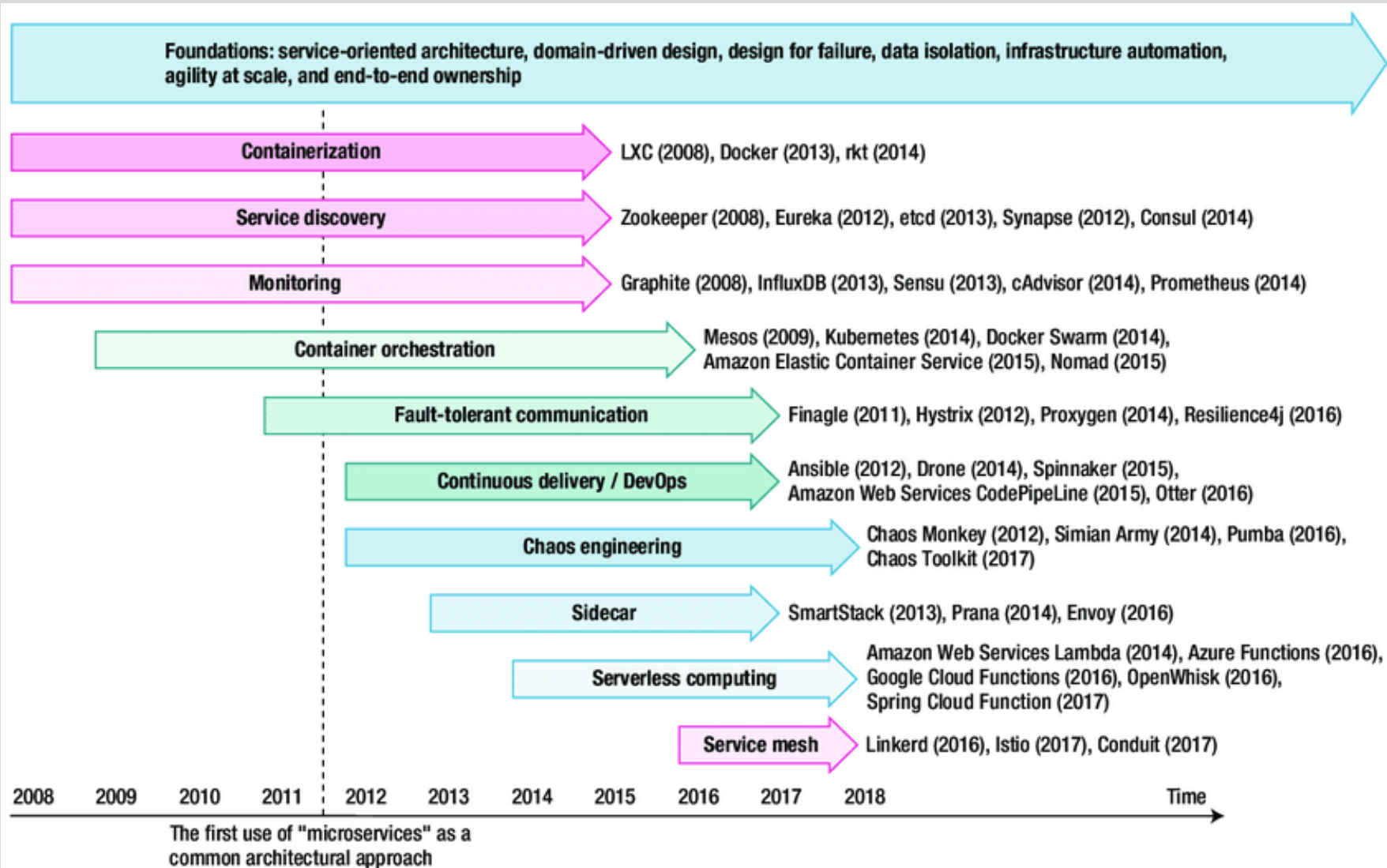
Cloud Native Readiness

Application Goals

Dai container al serverless



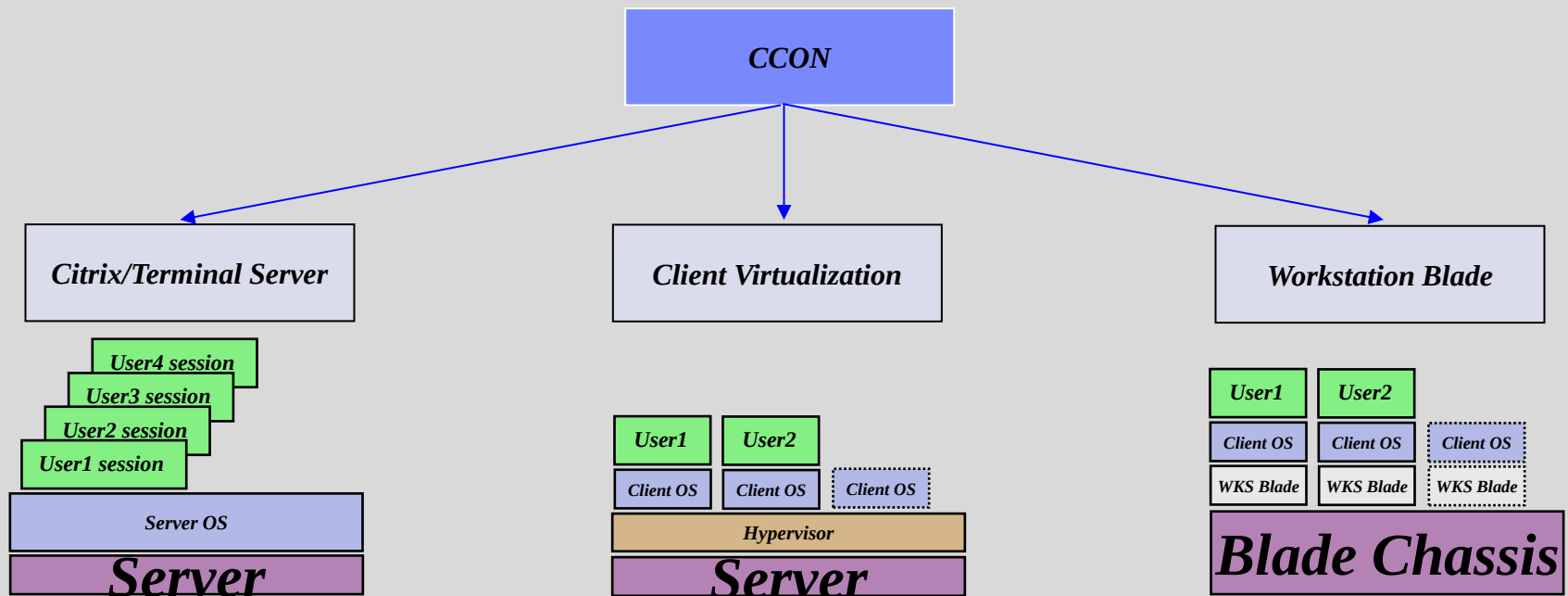
Modello micro-services



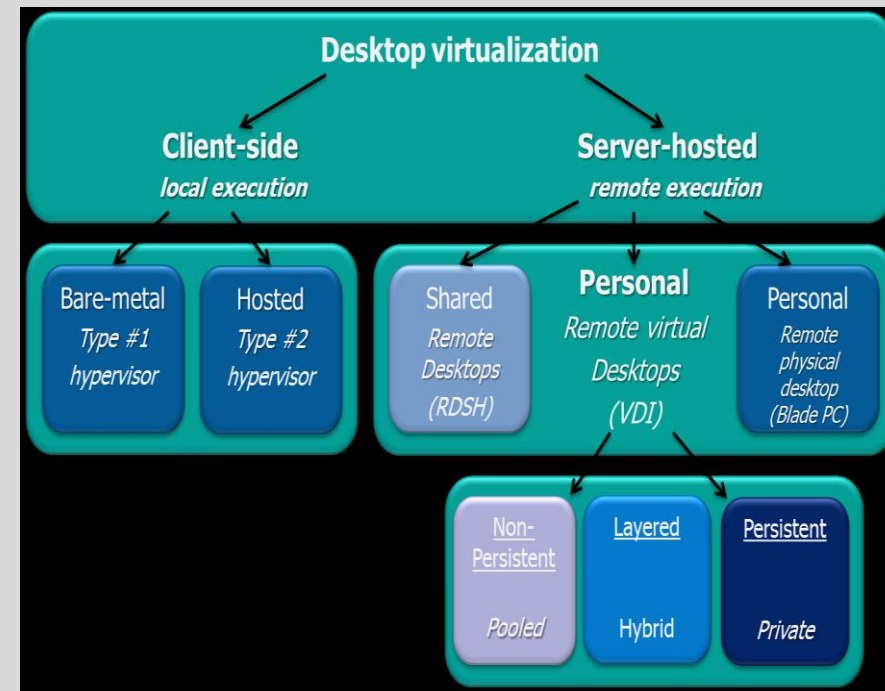
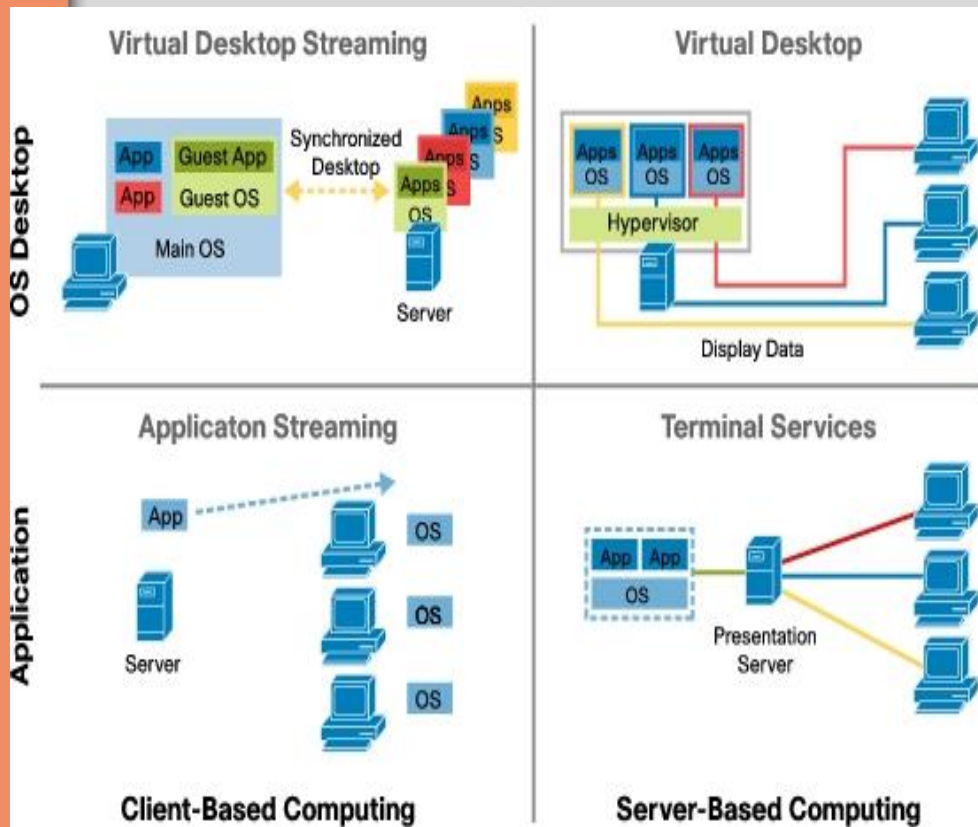
Virtualizzazione

VDI

CCON (Client Consolidation)



Differenti approcci



La strada verso il VDI

PC

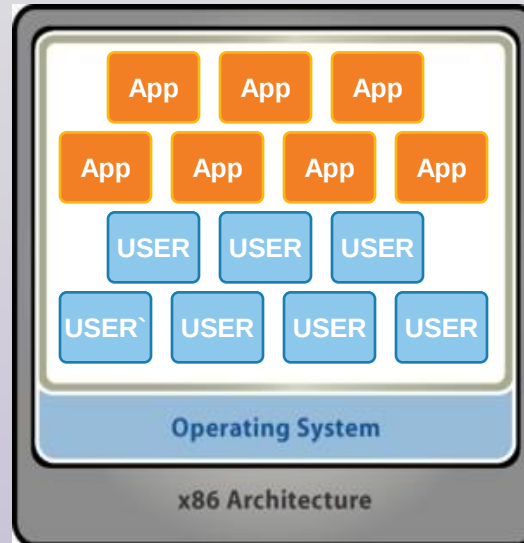


**Management, S/W,
H/W ???**

Support ???

Security ???

Terminal Server

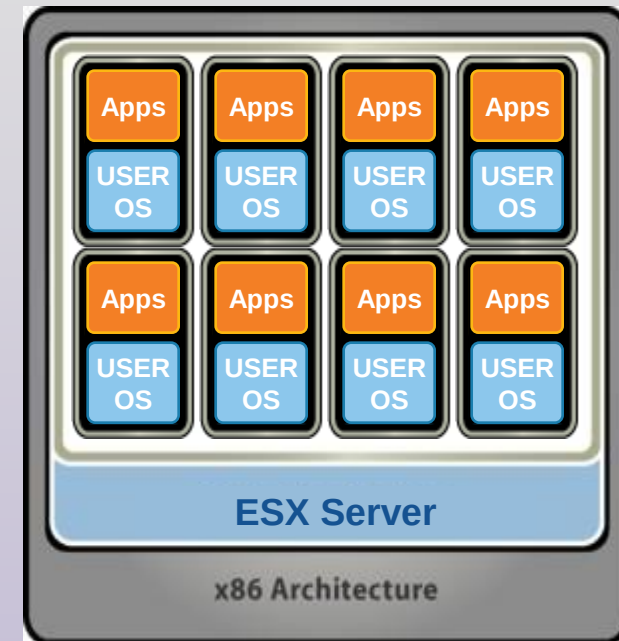


Gestione centralizzata

Isolamento?

Quali Applicazioni?

VDI

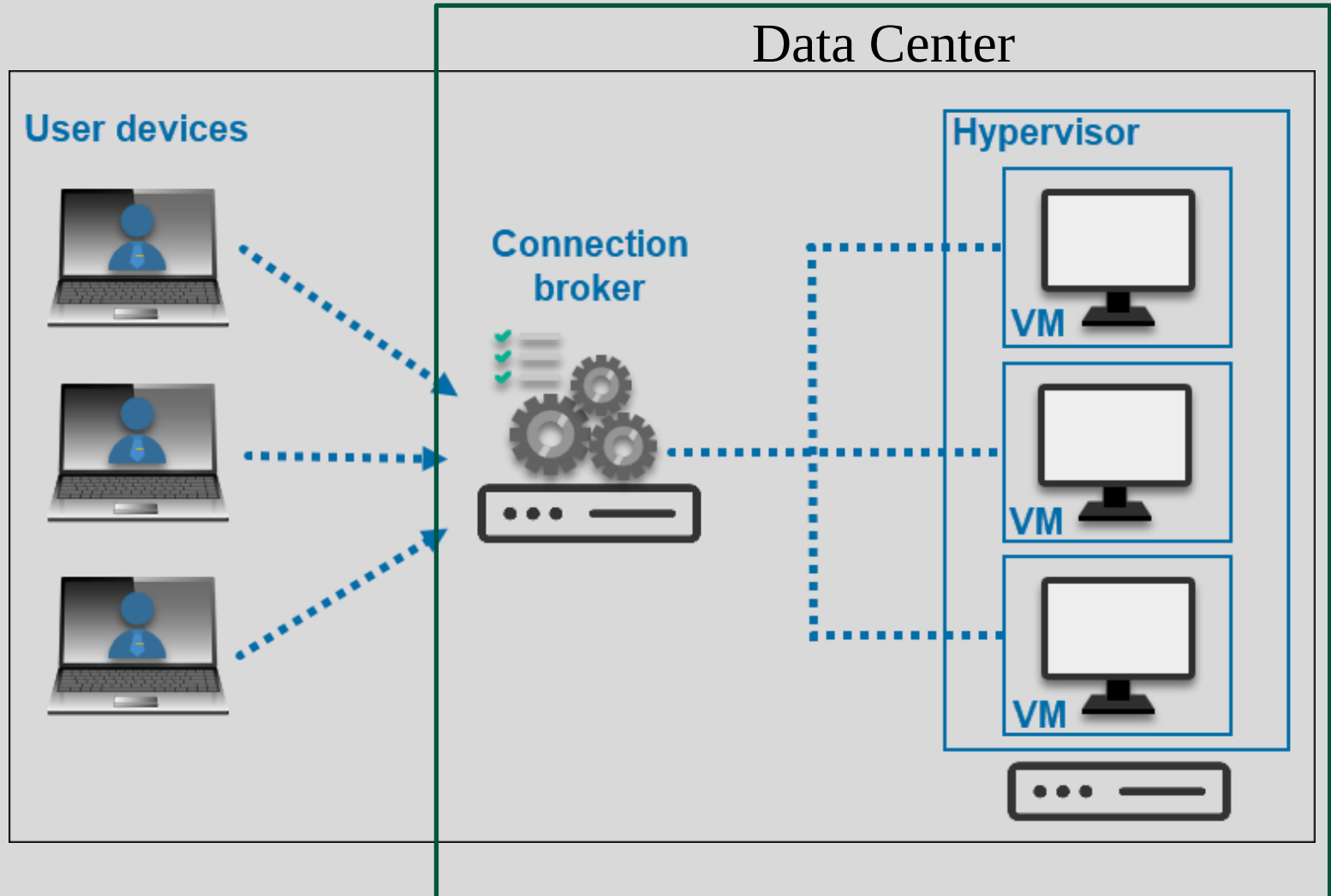


Gestione centralizzata

Qualunque applicazione

Isolamento

VDI



Virtual Desktop Infrastructure

Cloud native VDI	Cloud VDI	On-premises VDI
Image management	Image management	Image management
Directory services & policies	Directory services & policies	Directory services & policies
VDI control plane install & admin	VDI control plane install & admin	VDI control plane install & admin
Host admin	Host admin	Host admin
Storage admin	Storage admin	Storage admin
Load balancers install & admin	Load balancers install & admin	Load balancers install & admin
Hypervisor install & admin	Hypervisor install & admin	Hypervisor install & admin
Physical security	Physical security	Physical security
Power, HVAC	Power, HVAC	Power, HVAC
Rack and stack	Rack and stack	Rack and stack
		Customer managed
		Service managed

Soluzioni VDI

- On-prem
 - Citrix XenDesktop
 - VMware Horizon
 - Microsoft RDS
- Nel cloud
 - Solo control/management plane as a Service
 - Azure Virtual Desktop
 - Nutanix Frame
 - DaaS (Desktop as a Service)
 - Windows 365 (Cloud PC)
 - Amazon Workspace

IaaS, PaaS o SaaS?

	Windows 365 Business	Windows 365 Enterprise and Frontline	Azure Virtual Desktop
Cloud Service Model	Software as a Service	Software as a Service	Platform as a Service
Pricing Model	User subscription 300 user limit	User subscription No user limit, 3-to-1 ratio for Frontline	Consumption-based (pay as you go)
Costs	Predictable flat-rate •Per-user subscription license •Data cap set on vCPU count	Predictable flat-rate (optional Azure networking) •Per-user subscription license •Azure networking or data cap	Variable with component costs •Networking •Storage •Compute •Configuration & deployment
Management System	Online portal (optional Intune management)	Intune management (cloud or hybrid)	AVD native, Citrix, or VMWare
Management Complexity	Simple - Basic	Simple - Enterprise	Specialised
Skillset Required	No desktop skills needed	Commodity desktop skills	Specialised VDI skills
Operating Systems	Windows desktop single-session	Windows desktop single-session	Windows desktop & server single or multi-session
Desktop Experience	Personal persistent	Personal persistent	Personal, pooled, or app streaming

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